US 20250092433A1

(19) **United States**

(12) **Patent Application Publication** (10) **Pub. No.: US 2025/0092433 A1**

CASARETTO et al. (43) **Pub. Date: Mar. 20, 2025**

(54) **POLYPEPTIDES FOR USE IN THE SYNTHESIS OF KRATOM ALKALOIDS**

(71) Applicant: **Mitradyne Corporation**, Toronto (CA)

(72) Inventors: **Jose CASARETTO**, Guelph (CA); **Tariq A. AKTAR**, Guelph (CA); **Steven ROTHSTEIN**, Guelph (CA); **Eric SOUBEYRAND**, Kitchener (CA)

(73) Assignee: **Mitradyne Corporation**, Toronto (CA)

(21) Appl. No.: **18/891,491**

(22) Filed: **Sep. 20, 2024**

Related U.S. Application Data

(60) Provisional application No. 63/583,976, filed on Sep. 20, 2023.

Publication Classification

(51) **Int. Cl.**

C12P 17/18	(2006.01)
C12N 9/02	(2006.01)
C12N 9/10	(2006.01)
C12N 9/24	(2006.01)

(52) **U.S. Cl.**

CPC ***C12P 17/182*** (2013.01); ***C12N 9/001*** (2013.01); ***C12N 9/0073*** (2013.01); ***C12N 9/1007*** (2013.01); ***C12N 9/2402*** (2013.01); *C12Y 103/01036* (2013.01); *C12Y 114/13073* (2013.01); *C12Y 201/0105* (2013.01); *C12Y 201/01094* (2013.01); *C12Y 302/01105* (2013.01)

(57) **ABSTRACT**

Described herein are polypeptides encoding enzymes for the synthesis of monoterpene indole alkaloids. For example, the polypeptide comprises or consists of the sequence of SEQ ID NO: 1-68, or a variant thereof having at least 80% sequence identity to SEQ ID NO: 1-68, or a fragment of the polypeptide, or the variant thereof.

Specification includes a Sequence Listing.

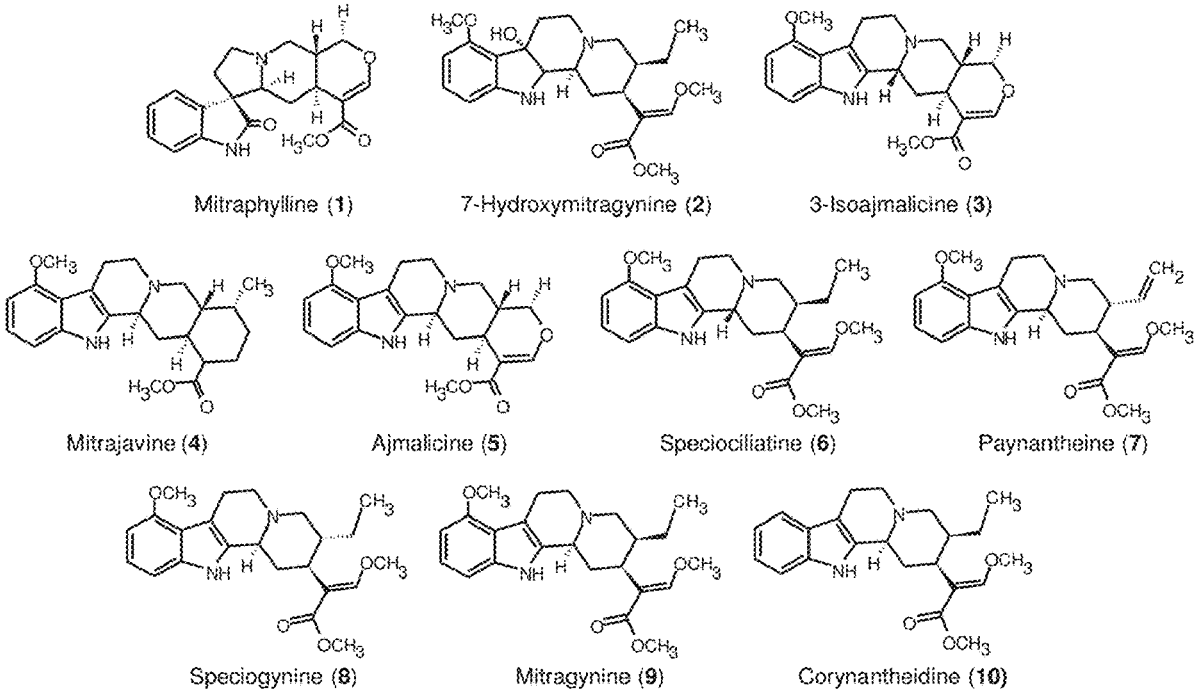


FIG. 1A

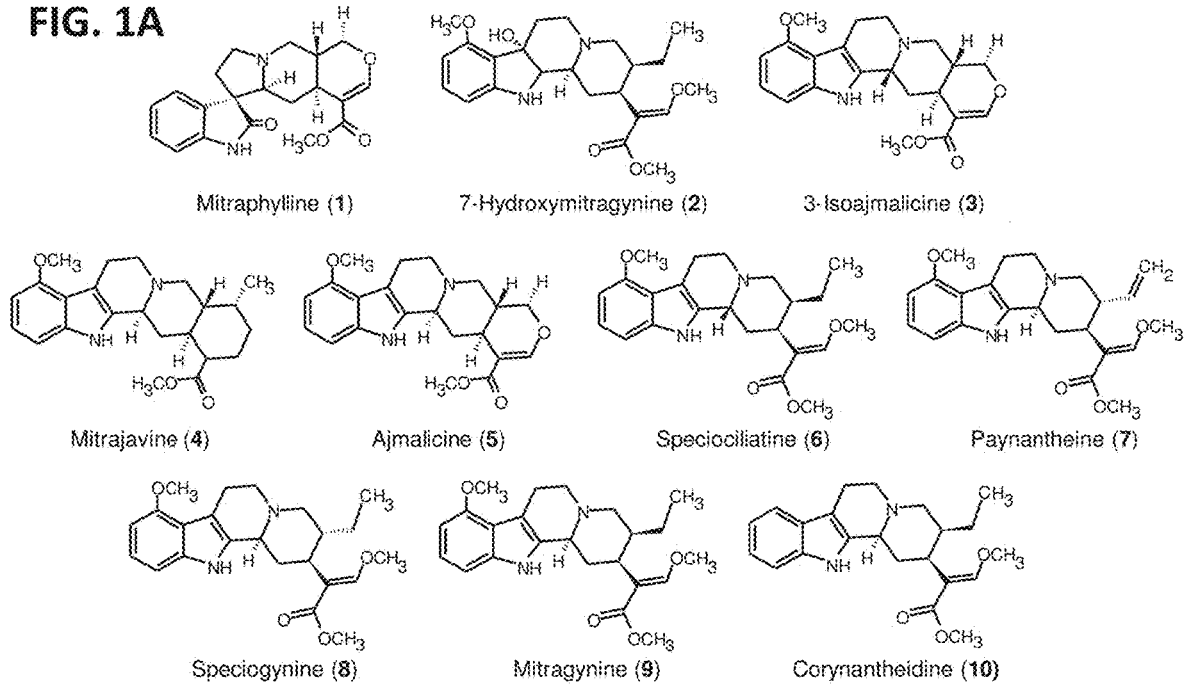


FIG. 1B

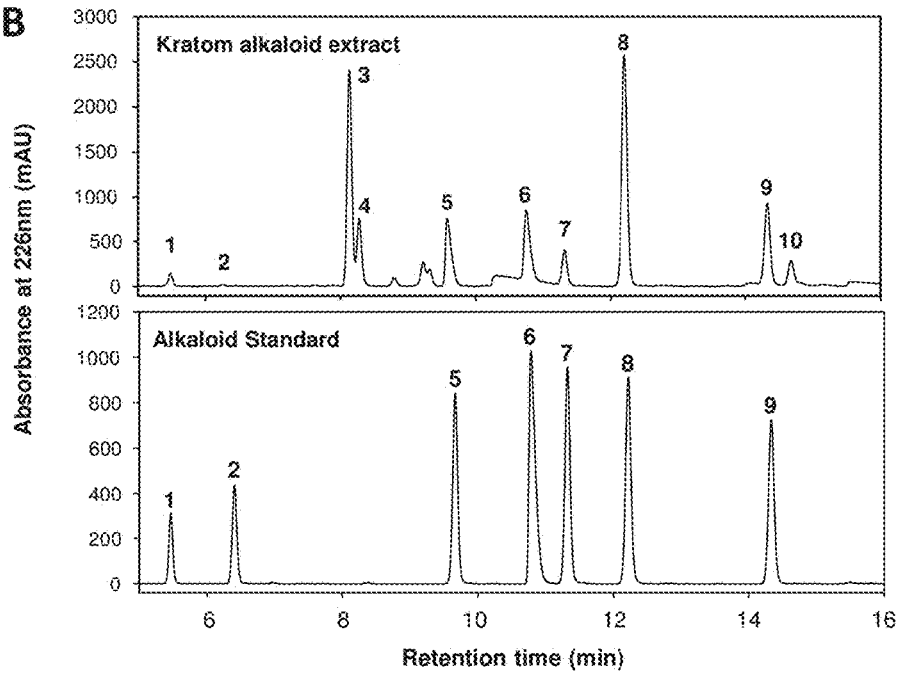


FIG 1. (Prior Art)

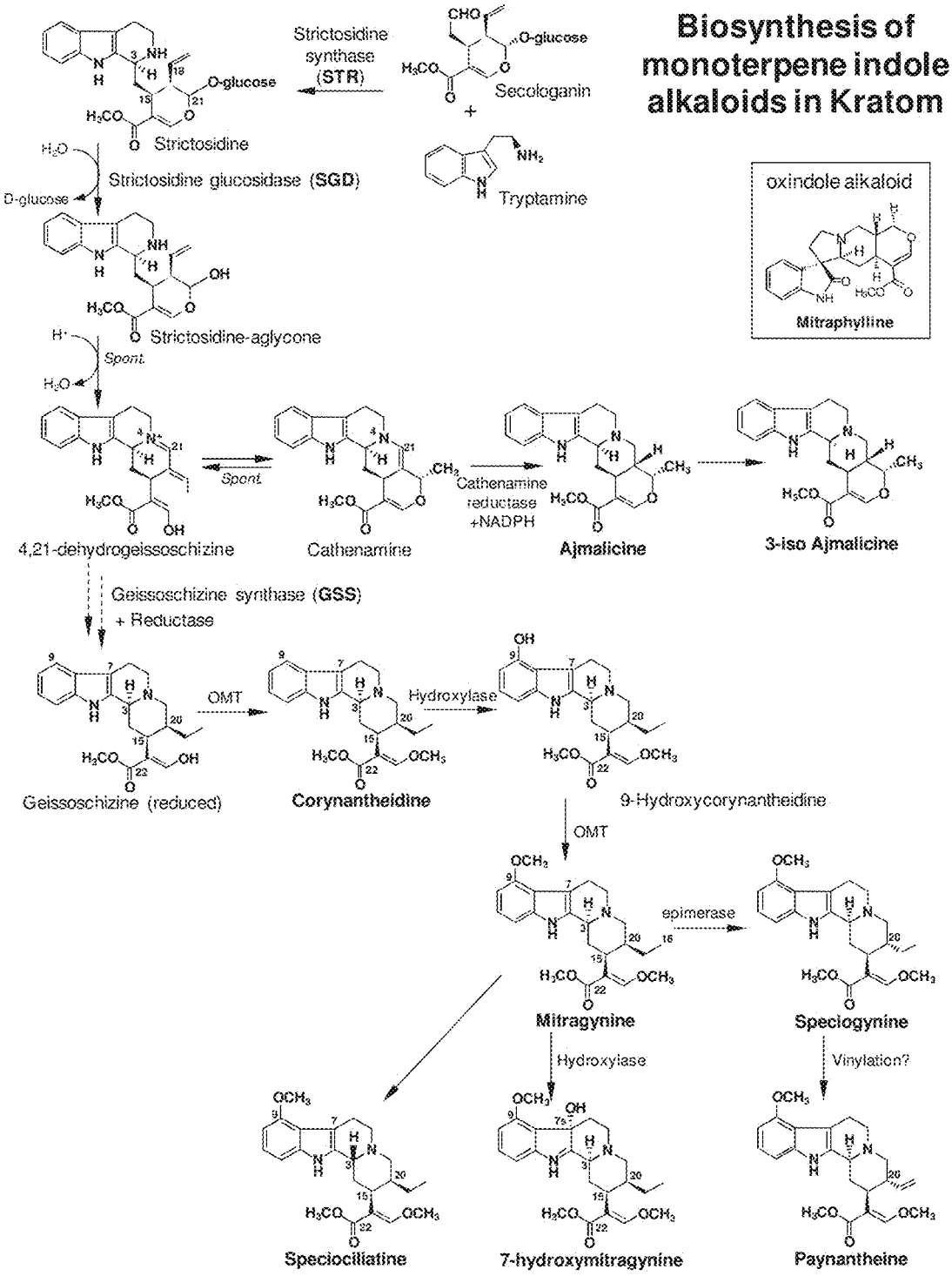

Biosynthesis of monoterpene indole alkaloids in Kratom
Strictosidine synthase (STR)
Secologanin
Tryptamine
Strictosidine
Strictosidine glucosidase (SGD)
Strictosidine-aglycone
Spont.
oxindole alkaloid
Mitraphylline
4,21-dehydrogeissoschizine
Cathenamine
Cathenamine reductase +NADPH
Ajmalicine
3-iso Ajmalicine
Geissoschizine synthase (GSS) + Reductase
OMT
Hydroxylase
Geissoschizine (reduced)
Corynantheidine
9-Hydroxycorynantheidine
epimerase
Mitragynine
Speciogynine
Vinylation?
Speciociliatine
7-hydroxymitragynine
Paynantheine

FIG. 2

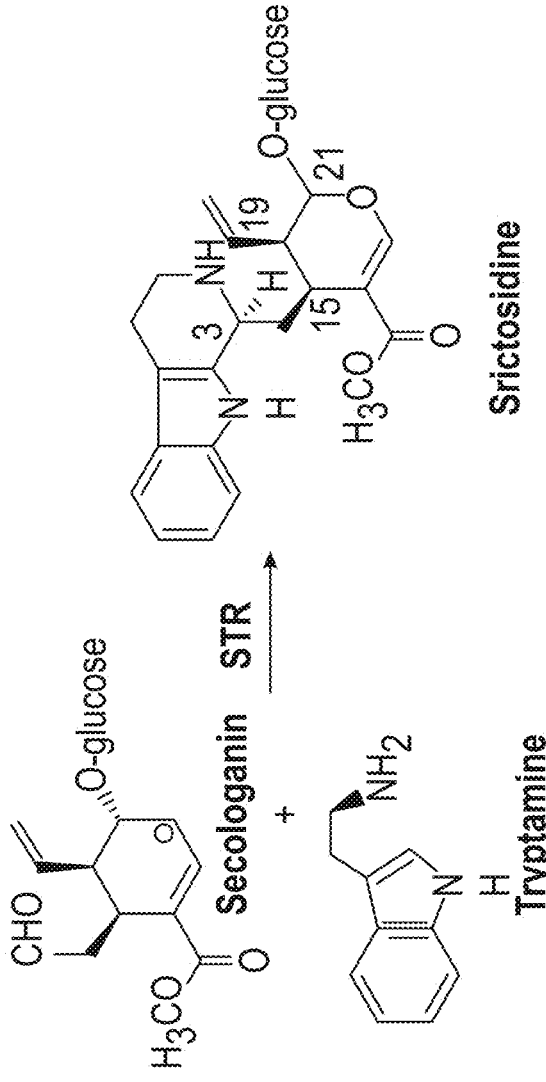

CHO
O-glucose
O
H_3CO
O
Secologanin
STR
+
NH_2
N
H
Tryptamine
O-glucose
21
O
19
NH
H
15
3
N
H
H_3CO
O
O
Srictosidine

FIG. 3A

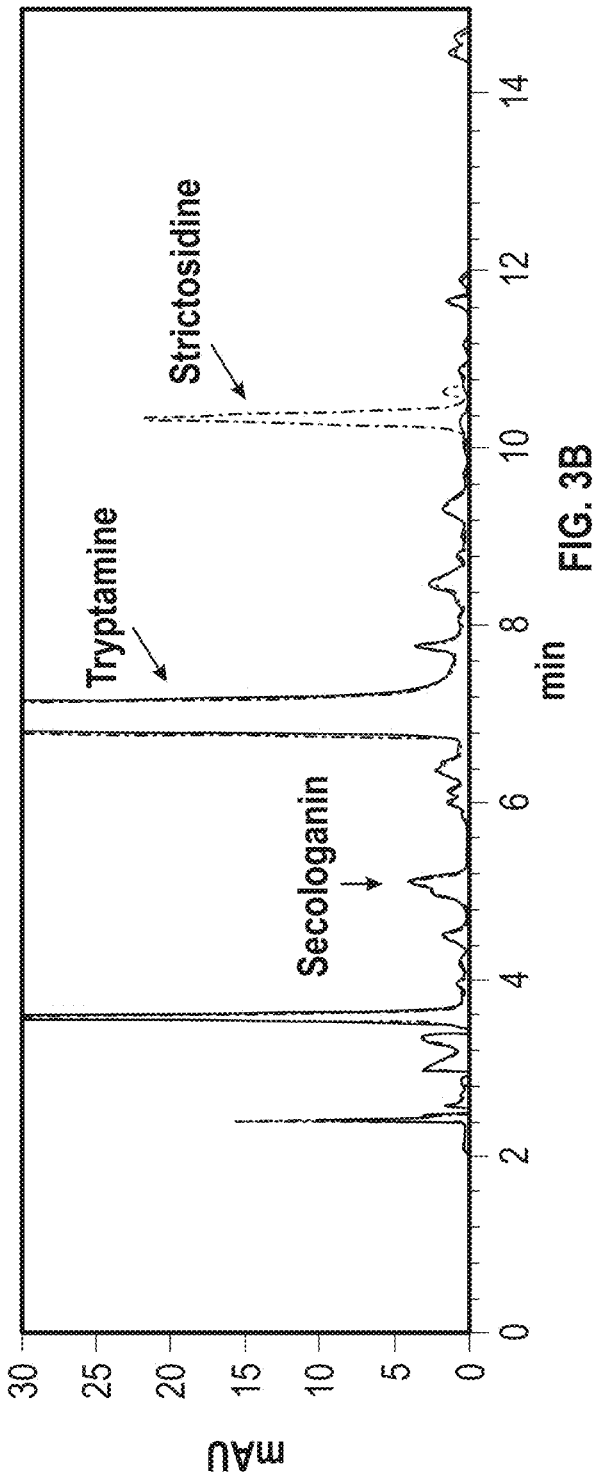
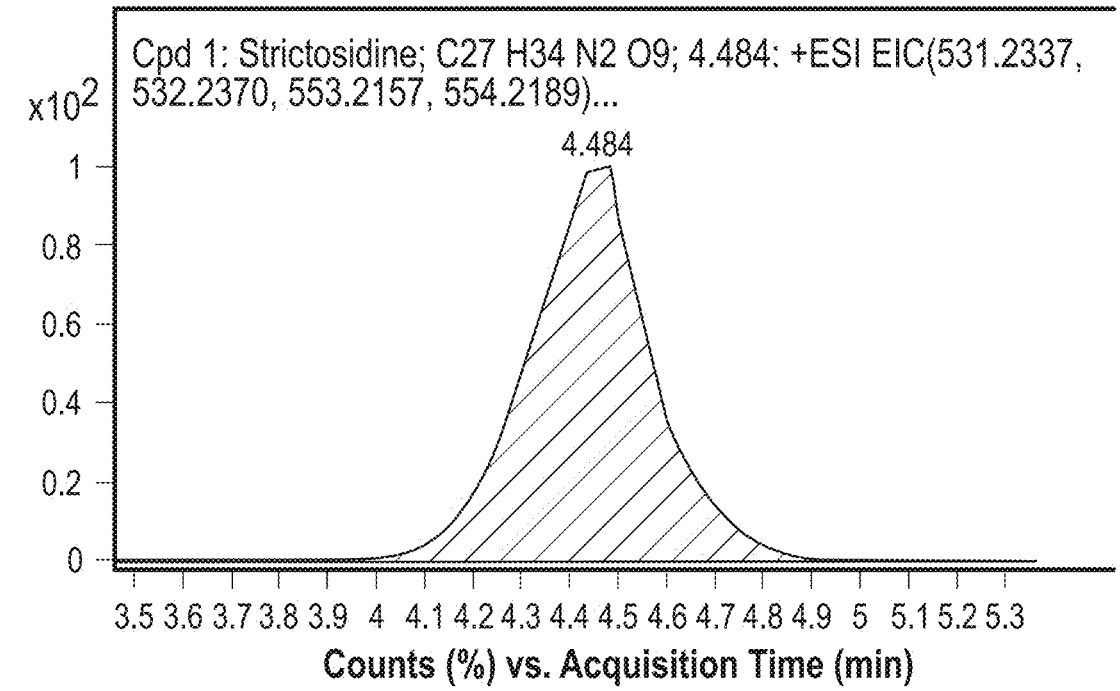

Tryptamine
Strictosidine
Secologanin
mAU
min
0
5
10
15
20
25
30
2
4
6
8
10
12
14

FIG. 3B


Cpd 1: Strictosidine; C27 H34 N2 O9; 4.484: +ESI EIC(531.2337, 532.2370, 553.2157, 554.2189)...
x10^2
4.484
1
0.8
0.6
0.4
0.2
0
3.5 3.6 3.7 3.8 3.9 4 4.1 4.2 4.3 4.4 4.5 4.6 4.7 4.8 4.9 5 5.1 5.2 5.3
Counts (%) vs. Acquisition Time (min)

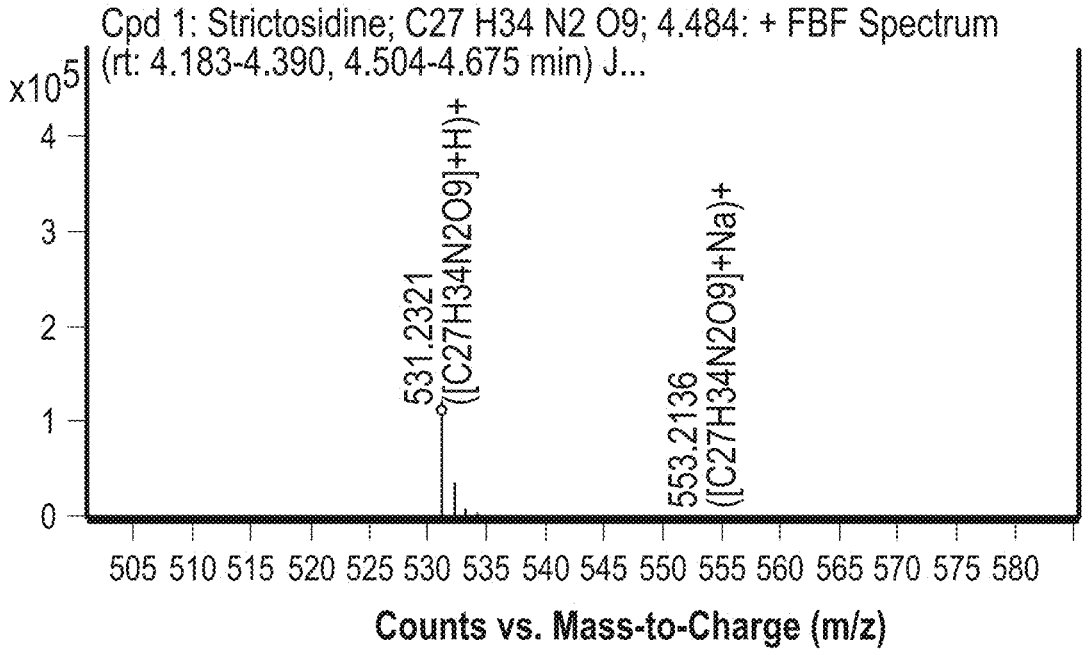

Cpd 1: Strictosidine; C27 H34 N2 O9; 4.484: + FBF Spectrum (rt: 4.183-4.390, 4.504-4.675 min) J...
x10^5
4
3
2
1
0
531.2321 ([C27H34N2O9]+H)+
553.2136 ([C27H34N2O9]+Na)+
505 510 515 520 525 530 535 540 545 550 555 560 565 570 575 580
Counts vs. Mass-to-Charge (m/z)

FIG. 3C

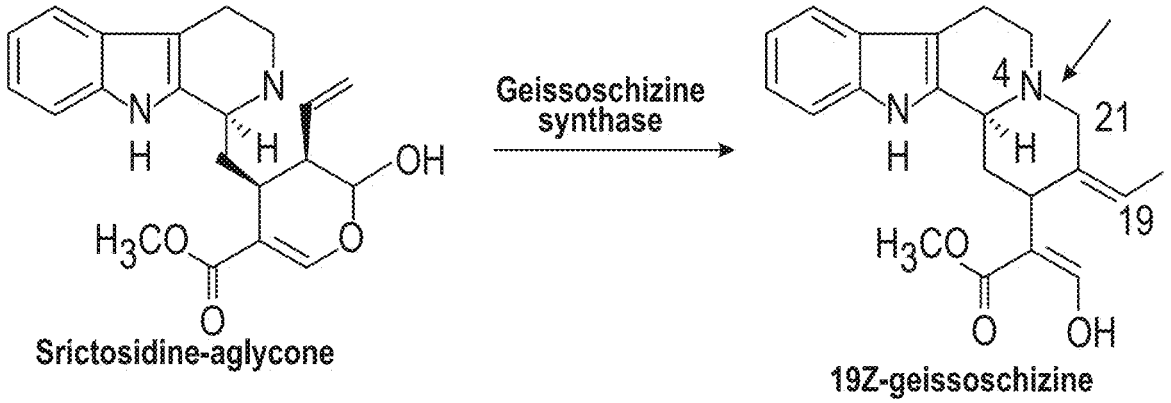

Geissoschizine synthase
Srictosidine-aglycone
19Z-geissoschizine

FIG. 4A

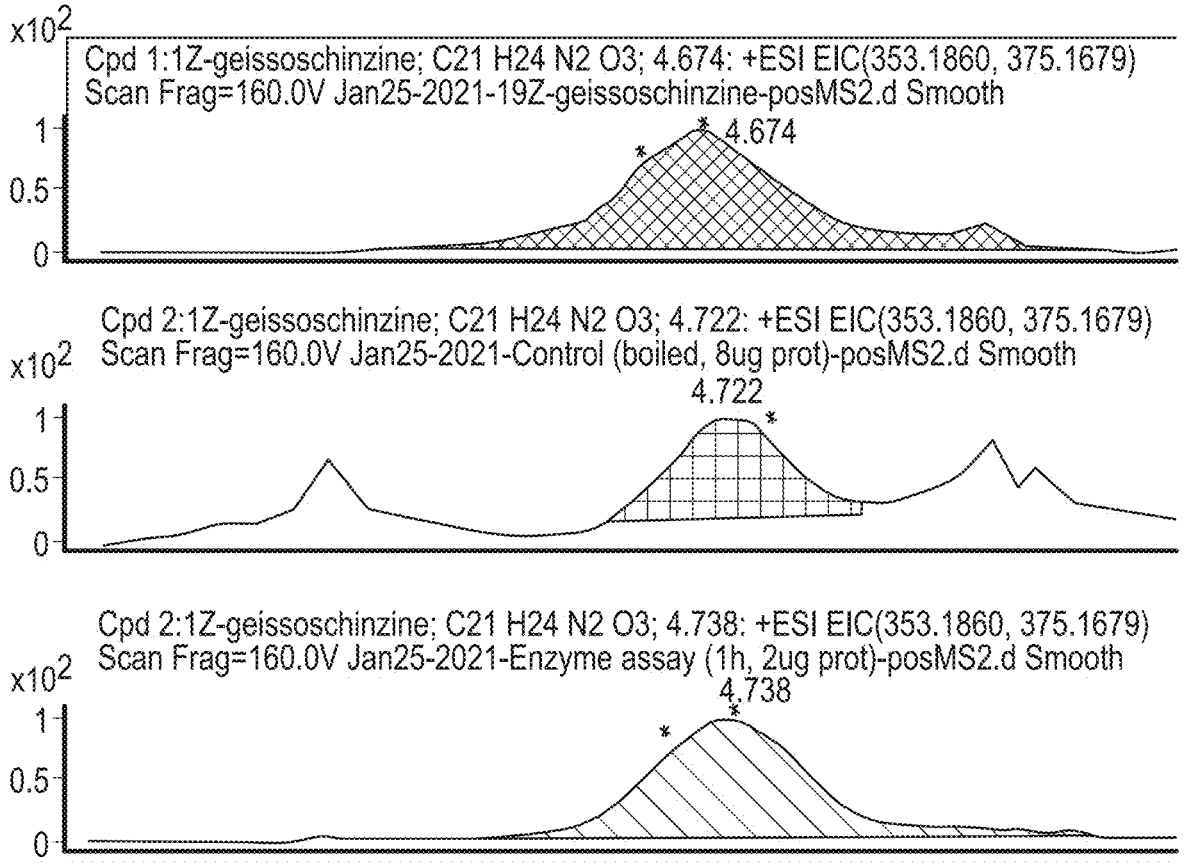

x10^2
Cpd 1:1Z-geissoschinzine; C21 H24 N2 O3; 4.674: +ESI EIC(353.1860, 375.1679)
Scan Frag=160.0V Jan25-2021-19Z-geissoschinzine-posMS2.d Smooth
4.674
Cpd 2:1Z-geissoschinzine; C21 H24 N2 O3; 4.722: +ESI EIC(353.1860, 375.1679)
Scan Frag=160.0V Jan25-2021-Control (boiled, 8ug prot)-posMS2.d Smooth
4.722
Cpd 2:1Z-geissoschinzine; C21 H24 N2 O3; 4.738: +ESI EIC(353.1860, 375.1679)
Scan Frag=160.0V Jan25-2021-Enzyme assay (1h, 2ug prot)-posMS2.d Smooth
4.738

FIG. 4B

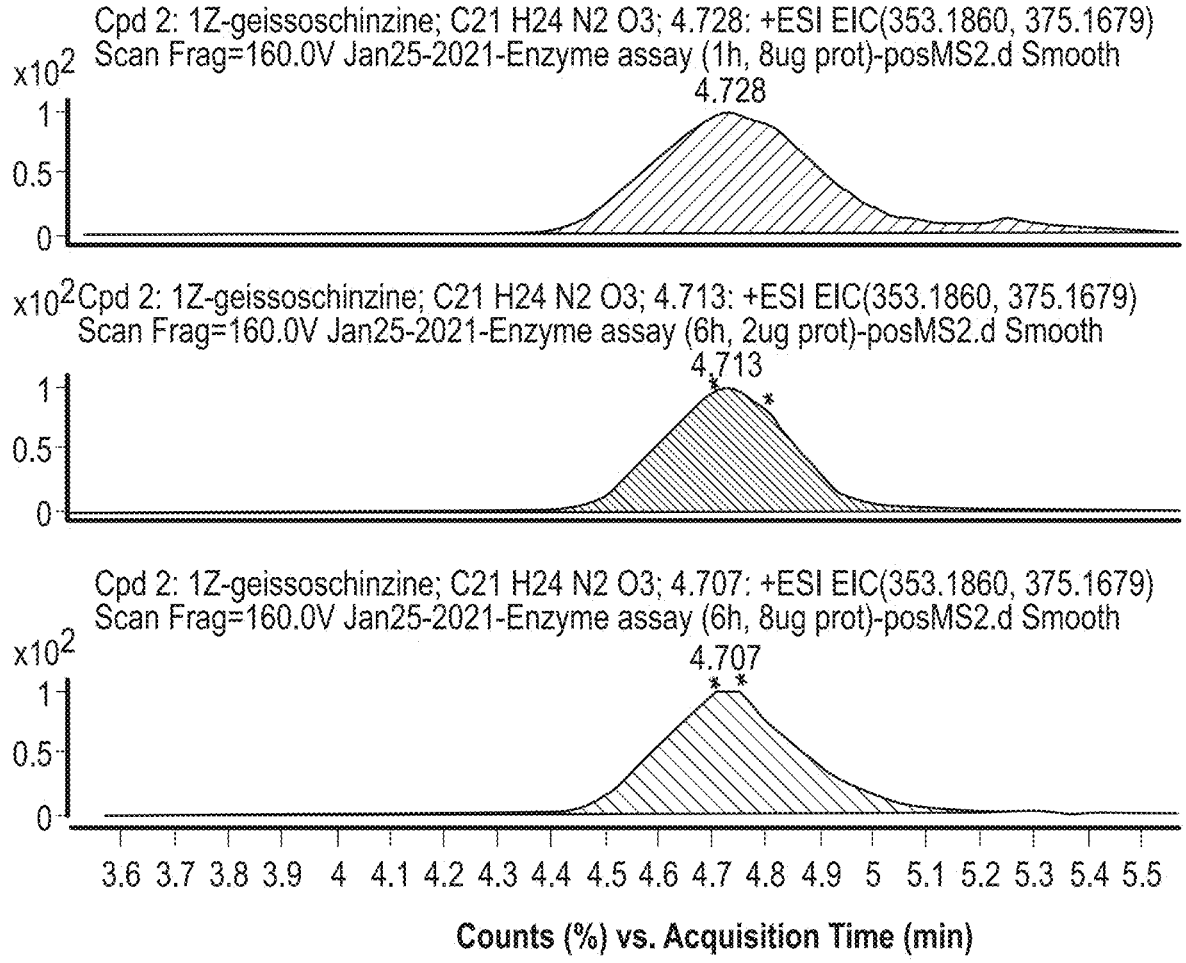


FIG. 4B
(Continued)

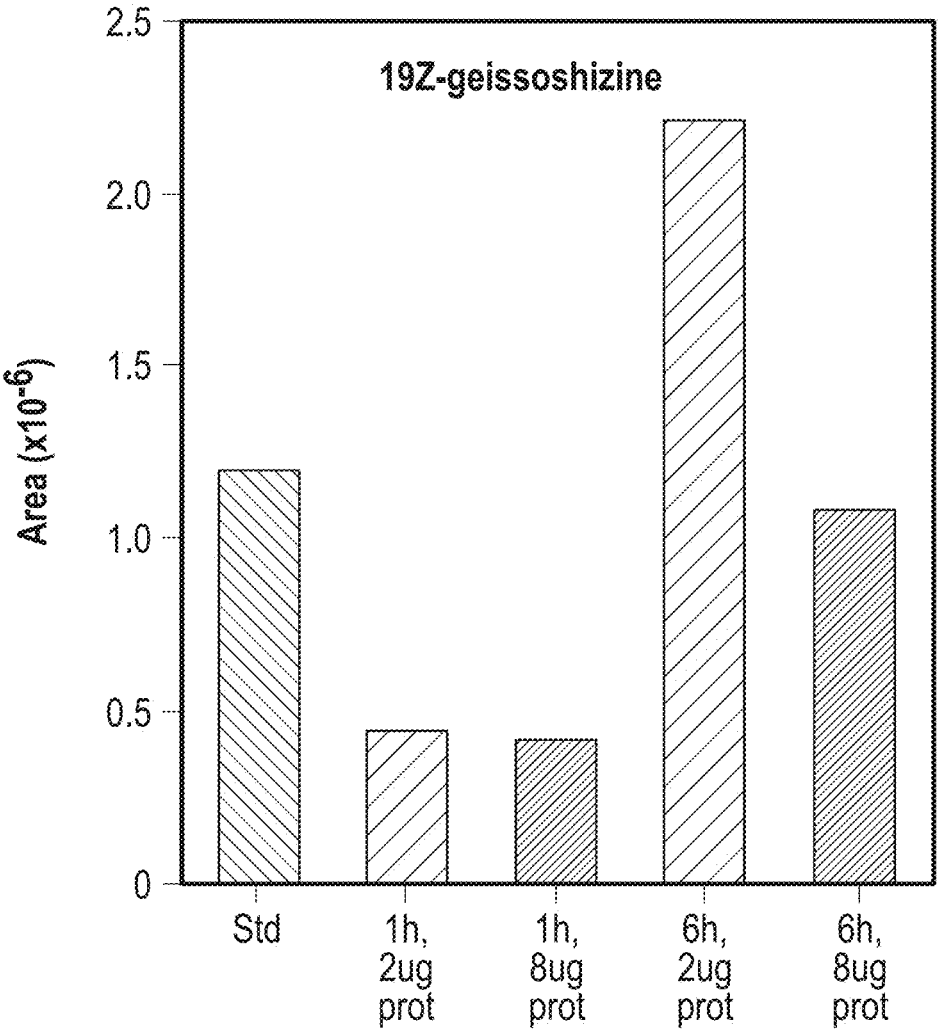

19Z-geissoshizine
Area (x10-6)
2.5
2.0
1.5
1.0
0.5
0
Std
1h, 2ug prot
1h, 8ug prot
6h, 2ug prot
6h, 8ug prot

FIG. 4C

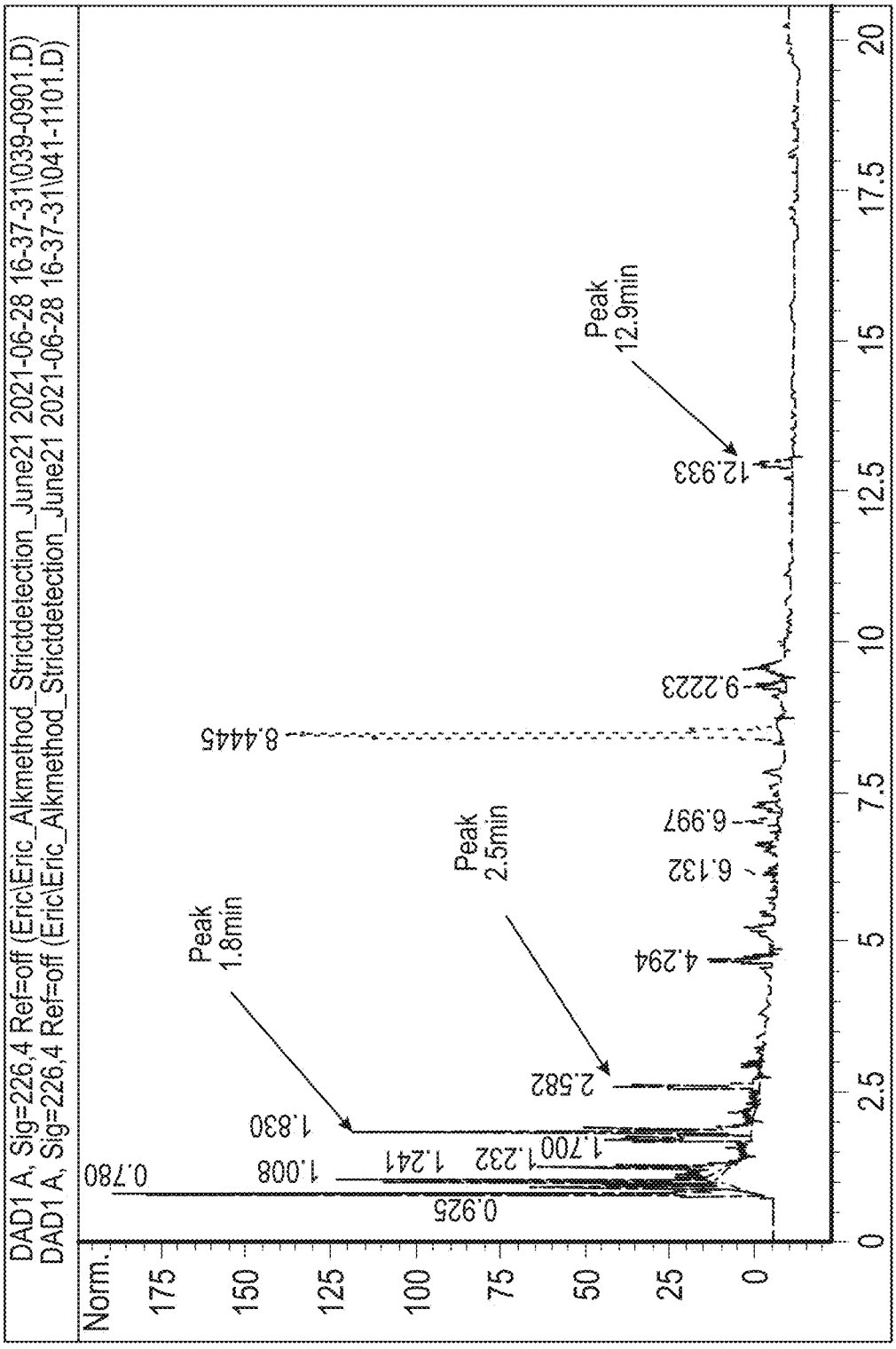
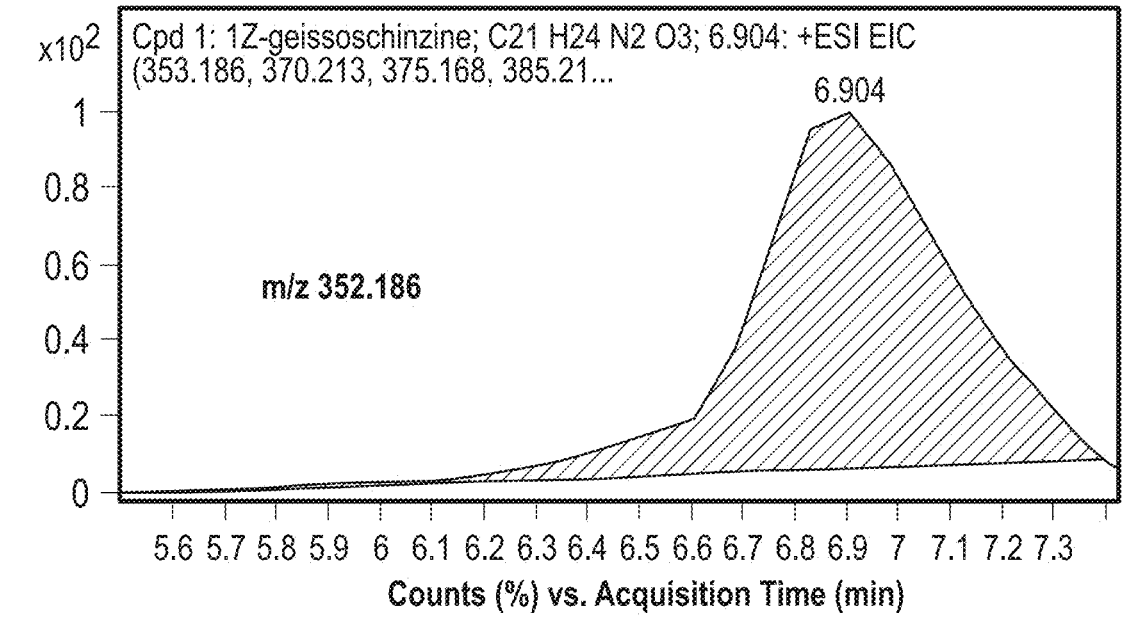

DAD1 A, Sig=226,4 Ref=off (Eric\Eric_Alkmethod_Strictdetection_June21 2021-06-28 16-37-31\039-0901.D)
DAD1 A, Sig=226,4 Ref=off (Eric\Eric_Alkmethod_Strictdetection_June21 2021-06-28 16-37-31\041-1101.D)
Norm.
175
150
125
100
75
50
25
0
0.780
0.925
1.008
1.232
1.241
1.700
1.830
2.582
4.294
6.132
6.997
8.445
9.223
12.933
Peak 1.8min
Peak 2.5min
Peak 12.9min
0
2.5
5
7.5
10
12.5
15
17.5
20

FIG. 5A


x10^2
Cpd 1: 1Z-geissoschinzine; C21 H24 N2 O3; 6.904: +ESI EIC
(353.186, 370.213, 375.168, 385.21...
6.904
m/z 352.186
1
0.8
0.6
0.4
0.2
0
5.6 5.7 5.8 5.9 6 6.1 6.2 6.3 6.4 6.5 6.6 6.7 6.8 6.9 7 7.1 7.2 7.3
Counts (%) vs. Acquisition Time (min)

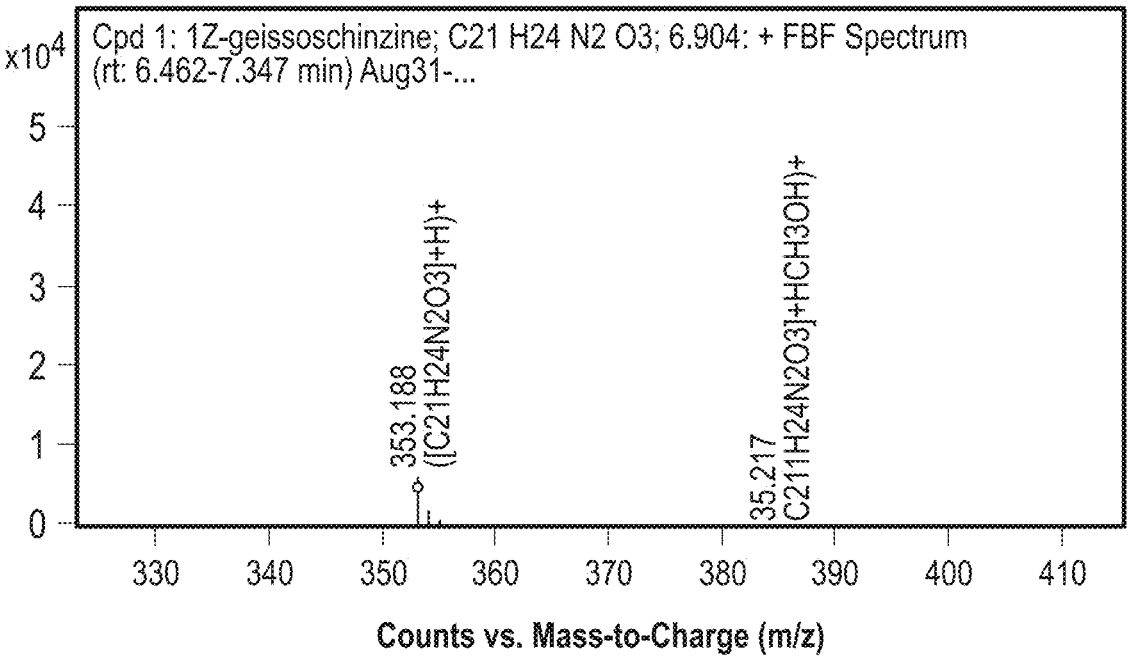

x10^4
Cpd 1: 1Z-geissoschinzine; C21 H24 N2 O3; 6.904: + FBF Spectrum
(rt: 6.462-7.347 min) Aug31-...
353.188
([C21H24N2O3]+H)+
385.217
[C211H24N2O3]+HCH3OH)+
5
4
3
2
1
0
330 340 350 360 370 380 390 400 410
Counts vs. Mass-to-Charge (m/z)

FIG. 5B

POLYPEPTIDES FOR USE IN THE SYNTHESIS OF KRATOM ALKALOIDS

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims priority under the Paris Convention to U.S. Provisional Patent Application No. 63/583,976 filed on Sep. 20, 2023, the contents of which are incorporated herein by reference.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

[0002] The Sequence Listing, created on 20 Sep. 2024, having a file size of 84 kilobyte and file name "2024.09.20_Sequence Listing_CPST_NP.xml" is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0003] The present invention relates to polypeptides. More specifically, the present invention is, in embodiments, concerned with the use of polypeptides encoding enzymes that can be used to synthesize indole alkaloid molecules and related products, methods, and uses.

BACKGROUND

[0004] Opioids are one of the most commonly used analgesic agents in clinical practice. These compounds bind specific receptors in the central and peripheral nervous system to elicit analgesia.

[0005] Naturally occurring alkaloids (plant-derived amines) isolated from opium poppy such as morphine, codeine, and papaverine are the most widely known plant-derived analgesics. In addition, chemical manipulations of these opiate alkaloids have produced a range of semi-synthetic opioids also useful in clinical medicine such as diamorphine, dihydrocodeine, buprenorphine, naloxone, and oxycodone. During the past decades, however, several synthetic opioids were also produced either by chance or by design. Examples of these synthetic compounds are butorphanol, levorphanol, tramadol, methadone, pentazocine, alfentanil, fentanyl, sufentanil, and remifentanil. Their prolonged use, misuse without medical supervision, or non-medical use, can lead to opioid dependence, and in overdose cases can lead to death. In recent years, about half a million deaths worldwide are attributable to drug use. In most countries, about 70% of these deaths are related to opioids, with more than 30% of those deaths caused by overdose (WHO, 2022; CDC, 2022: Gardner et al., 2022). There are, however, some treatments for opioid dependence that can decrease the risk of overdose. For example, naloxone, a competitive opioid receptor antagonist (binds to receptors but does not produce any functional response and prevents an agonist from binding to that receptor), can reverse the depression of the respiratory and central nervous systems caused by opioids and is used to prevent death from an opioid overdose if administered in time (CDC, 2022).

[0006] Individuals who suffer from chronic pain or want to reduce dependence on opioids and opiates are also focusing on other options. One of those is the plant known as kratom [*Mitragyna speciosa* (Korth.)], a tropical tree native to Southeast Asia (Coc et al., 2019; Palamar, 2021). In the last decade, its use has increased in popularity, especially in the U.S and Europe, although the medicinal use of kratom as a local anesthetic and substitute for opium in Malaysia and Thailand has been documented nearly 100 years ago (Field, 1921). Kratom has been traditionally consumed as a tea, or by chewing leaves, however, other formulations are currently available, including capsules and whole powder extracts. Kratom extracts have demonstrated opioid-receptor mediated analgesic effects when administered; whereas low doses have stimulatory effects, larger doses are associated with opioid-like sedative effects. This activity has been attributed to alkaloids to this plant exclusively, and in particular to mitragynine and 7-hydroxymitragynine (Prozialeck et al., 2012). Both compounds can bind to the human opioid receptors, and function as partial agonists of the μ-opioid receptor (μOR) and weak antagonists at κ-opioid and δ-opioid receptors (κOR, δOR) (Kruegel et al., 2016).

[0007] Unlike natural and clinically used synthetic opioids, kratom alkaloids do not promote β-arrestin recruitment to the opioid receptor when they activate the Gαi/o protein (Kruegel et al., 2016; Váradi et al., 2016). Several studies suggest that reduced recruitment of β-arrestin at the mu opioid receptors (μORs) and delta opioid receptor (δOR) is associated with weakened side effects caused by opioids such as respiratory depression (Gutridge et al., 2020; Mafi et al, 2020; Wilson et al., 2021). This mechanistic effect correlates with studies in animal models which have indicated that mitragynine has a low abuse potential (Hemby et al., 2019; Wilson et al., 2021). Nonetheless, this alkaloid can be metabolized into 7-hydroxymitragynine which exhibits greater potency than mitragynine and morphine (Avery et al., 2019; Kruegel and Grundmann 2018; Matsumoto et al., 2004), and is thought to be responsible for certain drug dependencies (Yusoff et al., 2016; Yue et al., 2018; Hemby et al., 2019; Gutridge et al., 2020). As a result, kratom has been subjected to the FDA's medical and safety evaluation and has prompted efforts to label 7-hydroxymitragynine and mitragynine Schedule I drugs by the Drug Enforcement Agency (Griffin and Webb, 2018). While not being illegal in the USA or most European countries, the possession and use of *Mitragyna speciosa* is controlled. In several countries of Europe, Southeast Asia, and Australia, kratom and/or mitragynine and 7-HMG are under control by narcotic laws due to their high misuse potential (Meireles et al., 2019).

[0008] Although mitragynine and 7-hydroxymitragynine have been the focus of multiple pharmacological investigations, kratom leaves are also able to produce 40 other alkaloids. The relative concentrations of such molecules depend on several factors including the variety of *Mitragyna speciosa*, (different varieties are found in different geographical region where the plant is grown), the harvest season, and the age of the leaves or plant (Adkins et al., 2011; Leon et al., 2009).

[0009] Kratom alkaloids are of the indole class of the corynanthe-type monoterpenoid alkaloids and oxindole alkaloids (Flores-Bocanegra et al., 2020). Besides the two already mentioned, other examples of such components are: paynantheine, speciociliatine, speciogynine, corynantheidine, corynoxine, corynoxine B, isocorynantheidine, mitraphilline (Sharma et al., 2019; FIG. 1). In addition, other compounds, such as flavonoids, terpenoids, and polyphenols have also been identified (Feng et al., 2017; Vecramohan et al., 2018).

[0010] The limited pharmacological data currently available for most of the kratom alkaloids warrants the investigation of pure alkaloids present in the plant and their

significance in different products used medicinally. Several considerations indicate the need for more research on these molecules. One fact is that there exist differences in the genetics of kratom plants (e.g. different varieties) that influence the phytochemistry of the plant, hence, the same alkaloids are not present in the same proportion. Kratom products are sold under a variety of names, however, there is no information regarding the chemical composition of these preparations. Therefore there is a need for standardized formulations, especially if they are for medicinal use. For example, the relative potency of some kratom alkaloids on the opioid receptors is still unknown, therefore is difficult to predict the efficacy, reduced side effects, and potential health implications of products containing them.

[0011] The biosynthesis of the indole alkaloids relies on two precursors, tryptamine which is derived from the amino acid tryptophan providing the indol moiety to the structure, and secologanin, an iridoid monoterpene (Charoonratana et al., 2013). The biosynthesis pathway starts with the condensation reaction of tryptamine with secologanin to form strictosidine (Jumali et al., 2011), a precursor of hundreds of derivatives of monoterpene indole alkaloids in several plant families (Treimer & Zenk, 1979; de Luca et al., 2014). Geissoschizine and cathenamine are two examples of monoterpene alkaloids that can be synthesized starting from strictosidine. These alkaloids have been identified in fewer species, including kratom (de Luca et al., 2014; Sharma et al., 2019). Therefore, while the synthesis of indole alkaloids branches out from strictosidine to the diversification of these molecules among many plant species, those produced downstream in the biosynthetic pathway in *M. speciosa* are exclusive to this plant.

[0012] Recently, two enzymatic steps of intermediate molecules for the synthesis of mitragynine were described (Schotte et al., 2023), however all the enzyme-encoding genes associated with the synthesis pathway of indole alkaloids in kratom is still elusive.

[0013] Various kratom alkaloids and their use have been previously published as in the following examples.

[0014] U.S. Publication No. 8648090B2 describes indole alkaloid derivatives having an opioid receptor agonistic effect, their synthesis, and therapeutic compositions containing these derivatives, and methods of treating conditions with these compounds and therapeutic compositions, are provided.

[0015] U.S. Patent Publication U.S. Pat. No. 10,751,380B2 relates to pharmaceutical and nutraceutical compounds and methods for reducing inflammation, muscle spasms, and pain associated with cancer, trauma, medical procedure, neurological diseases, and disorders, as well as other conditions in subjects in need thereof. The disclosed invention further relates to *Mitragyna speciosa* and *Cannabis sativa* based compounds and synthetic analogs of the alkaloids found in said plants, where said compounds can also contain a pharmacologically inactive substance for modifying certain qualities of said compounds.

[0016] U.S. Patent Publications U.S. Pat. No. 10,961,244B2 and WO2017165738A1 relate to a compound having a structure related to mitragynine alkaloids or a pharmaceutically acceptable salt or ester, and a method of treating a subject afflicted with pain, a depressive disorder, a mood disorder or an anxiety disorder by administering the compound to the subject.

[0017] U.S. Patent Applications No. 2021/0113644A1 and WO2021076849A1 disclose compositions containing two purified kratom compounds chosen from the following: 7-Hydroxymitragynine, Mitragynine, Paynantheine, Speciociliatine, Speciogynine, Ajmalicine, Ciliaphylline, Corynantheidine, Corynoxine A, Corynoxine B, Isomitraphylline, Isorhynchophylline, Mitraphylline, Rhynchophylline, Speciophylline, Speciofoline, Epicatechin, 7-Hydroxyspecioliatine, 9-Hydroxycorynantheidine, Corynoxeine, Isopteropodinc, Isorhynchophylline Oxindole, Tetrahydroalstonine, Mitragynine Oxindole B, and Mitragynine Oxindole A or the salts of these kratom compounds. The disclosure also relates to formulations, including pharmaceutical formulations, of such a composition and an excipient. Pharmaceutical formulation further containing a therapeutically effective amount of a serotonergic drug, a purified psilocybin derivative, a purified cannabinoid, or a purified terpene are disclosed. Also disclosed are methods of regulating the activity of a neurotransmitter receptor and methods of treating a psychological disorder, a compulsive disorder, or a depressive disorder.

[0018] U.S. Patent Publication No. 11046692B2 describes compounds formulae of mitragynine analogs and pharmaceutically acceptable derived salts that are useful for modulating opioid receptor activity. The provided compounds may have both agonistic and antagonistic effects on one or more opioid receptors. Methods of using the compounds for treating or managing pain are also described.

[0019] U.S. patent Application No. 20220135564A1 describes a compound having the structure of a deuterated mitragynine or a pharmaceutically acceptable salt or ester thereof, and methods of using the compound to treat pain, depressive disorders, mood disorders, anxiety disorders, opioid use disorder, and opioid withdrawal symptoms.

[0020] U.S. patent Application No. 20210179619A1 relates to mitragynine analogs for the treatment of pain, mood disorders and substance use disorders. The invention provides a compound having the structure of an indole alkaloid related to mitragynine, and a method of treating a subject afflicted with pain, a depressive disorder, a mood disorder or an anxiety disorder by administering the compound to the subject.

[0021] Patent Publication WO2022008955 describes preparations containing Mitragyne extract or its isolated alkaloids and *cannabis* extract or its isolated cannabinoids, and cosmetic and/or pharmaceutical preparation for topical use containing from about 0.00001 wt.-% to about 20 wt.-%, of one or more alkaloids of *Mitragyna speciosa* and from about 0.00001 wt.-% to about 20 wt.-% of one or more cannabinoids of *Cannabis* indica or *Cannabis sativa*. The preparation is intended for use in preventing and/or treating skin and mucosal disorders and conditions.

[0022] Patent Publication CN112094265 describes an extraction method of 7-OH-mitragynine, which comprises the following steps: (1) pulverizing dried kratom leaves, adding an extractant, carrying out ultrasonic extraction at normal temperature, and carrying out centrifugation and vacuum concentration to obtain a first concentrated solution; (2) adding an extraction agent into the first concentrated solution for extraction, and performing vacuum concentration to obtain a second concentrated solution; and (3) carrying out silica gel column chromatography separation on

the second concentrated solution, and carrying out gradient elution with an eluent to obtain the 7-OH-mitragynine with high purity.

[0023] Research and potential utilization of individual or specific combinations of kratom alkaloids are impeded by the lack of understanding of their biosynthesis in the kratom plant. Therefore, there is a need to provide a useful alternative to overcome at least some of the deficiencies of the prior art.

SUMMARY

[0024] In accordance with an aspect, there is provided a polypeptide encoding a strictosidine glucosidase that removes the glucose moiety from strictosidine to produce strictosidine aglycone.

[0025] In an aspect, the polypeptide is a strictosidine glucosidase from *Mitragyna speciosa*.

[0026] In an aspect, the polypeptide is comprising or consisting of a polypeptide having at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the sequence of any one or more of SEQ ID NO: 1-13, or a fragment of any thereof.

[0027] In an aspect, the polypeptide produces strictosidine-aglycone when reacted under suitable conditions in the presence of strictosidine.

[0028] In accordance with an aspect, there is provided a polypeptide encoding a geissoschizine synthase that opens the tetrahydropyran ring of strictosidine aglycone to produce geissoschizine.

[0029] In an aspect, the polypeptide is a geissoschizine synthase from *Mitragyna speciosa*.

[0030] In an aspect, the polypeptide is comprising or consisting of a polypeptide having at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the sequence of any one or more of SEQ ID NO: 14-22, or a fragment of any thereof.

[0031] In an aspect, the polypeptide produces geissoschizine when reacted under suitable conditions in the presence of strictosidine aglycone.

[0032] In accordance with an aspect, there is provided a polypeptide encoding a hydroxylase that hydroxylates the indole ring of corynantheidine at the C9 position to produce 9-hydroxycorynantheidine.

[0033] In accordance with an aspect, the polypeptide encoding a hydroxylase hydroxylates the indole ring of mitragynine at the C7 position to produce 7-hydroxymitragynine.

[0034] In an aspect, the polypeptide is a hydroxylase from *Mitragyna speciosa*.

[0035] In an aspect, the polypeptide is comprising or consisting of a polypeptide having at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the sequence of any one or more of SEQ ID NO: 23-54, or a fragment of any thereof.

[0036] In an aspect, the polypeptide produces a 9-hydroxy-indole alkaloid, such as 9-hydroxycorynantheidine.

[0037] In an aspect, the polypeptide produces 7-hydroxymitragynine when reacted under suitable conditions in the presence of mitragynine.

[0038] In accordance with an aspect, there is provided a polypeptide encoding an O-methyltransferase that methylates the hydroxyl group at the C9 position of 9-hydroxycorynantheidine to produce mitragynine.

[0039] In an aspect, the polypeptide is an O-methyltransferase from *Mitragyna speciosa*.

[0040] In an aspect, the polypeptide is comprising or consisting of a polypeptide having at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the sequence of any one or more of SEQ ID NO: 55-65, or a fragment of any thereof.

[0041] In an aspect, the polypeptide produces mitragynine when reacted under suitable conditions in the presence of 9-hydroxycorynantheidine.

[0042] In an aspect, the polypeptide, variant, or fragment comprises up to about 100, about 150, about 200, about 250, about 300, about 350, about 360, about 370, about 380, about 390, about 400, about 410, about 420, about 430, about 440, about 450, or about 500 amino acids

[0043] In an aspect, the polypeptide is synthetic.

[0044] In an aspect, the polypeptide is recombinant.

[0045] In accordance with an aspect, there is provided a nucleic acid encoding the polypeptide described herein.

[0046] In an aspect, the nucleic acid is cDNA.

[0047] In accordance with an aspect, there is provided a vector comprising the nucleic acid described herein.

[0048] In accordance with an aspect, there is provided a host cell comprising the vector described herein.

[0049] In accordance with an aspect, there is provided a host cell expressing the polypeptide described herein.

[0050] In an aspect, the host cell is a bacterial cell (e.g., *E. coli* or *Agrobacterium tumefaciens*), a yeast cell (e.g., *S. cerevisiae*), an algal cell, or a plant cell (e.g., *Nicotiana* spp.).

[0051] In an aspect, the host cell is provided in combination with an indole alkaloid precursor.

[0052] In an aspect, the indole alkaloid precursor is provided in the host cell culture medium.

[0053] In an aspect, the indole alkaloid precursor is expressed by the host cell.

[0054] In an aspect, the indole alkaloid precursor is strictosidine.

[0055] In accordance with an aspect, there is provided an expression system comprising the polypeptide; the nucleic acid, the vector; or the host cell described herein.

[0056] In accordance with an aspect, there is provided a system for modifying an indole alkaloid the system comprising the polypeptide described herein.

[0057] In an aspect, the system is cell-free.

[0058] In accordance with an aspect, there is provided a method for modifying an indole alkaloid, wherein the method comprises contacting the indole alkaloid with a polypeptide described herein.

[0059] In an aspect, the method is a recombinant method comprising expressing a polypeptide described herein in a cell in the presence of an indole alkaloid precursor.

[0060] In an aspect, the method comprises combined enzymatic reaction steps.

[0061] In accordance with an aspect, there is provided a method of producing strictosidine aglycone, 4,21-dehydrogeissoschizine, geissoschizine, corynantheidine, 9-hydroxycorynantheidine, mitragynine, 7-hydroxymitragynine, speciociliatine, speciogynine, and paynantheine, the method comprising carrying out combined enzymatic reactions with the polypeptides described herein.

[0062] In an aspect, the final indole alkaloid is substantially pure, for example, at least about 95%, about 96%, about 97%, about 98%, about 99%, about 99.5%, or about 99.9% pure.

[0063] In accordance with an aspect, there is provided a pharmaceutical composition comprising the indole alkaloid described herein and at least one pharmaceutically acceptable carrier.

[0064] In accordance with an aspect, there is provided a natural health product comprising the indole alkaloid described herein, such as a supplement, beverage, or food.

[0065] In accordance with an aspect, there is provided a use of the indole alkaloid described herein in a pharmaceutical or natural health product.

[0066] Other features and advantages of the present invention will become apparent from the following detailed description. It should be understood, however, that the detailed description and the specific examples while indicating embodiments of the invention are given by way of illustration only since various changes and modifications within the spirit and scope of the invention will become apparent to those skilled in the art from the detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0067] The present invention will be further understood from the following description concerning the Figures:

[0068] FIG. **1** depicts alkaloids detected in a crude kratom leaf extract (prior art). (FIG. **1**A) Chemical structure of alkaloids detected in the extracts. (FIG. **1**B) Representative HPLC chromatogram illustrating the alkaloid profile of a final crude extract and detected at 226 nm. The different peaks (top panel) were compared with available standards (bottom panel) and then collected and analyzed via mass spectrometry. The peaks represent, respectively: 1, mitraphylline; 2, 7-hydroxymitragynine; 3, 3-isoajmalicine; 4, mitrajavine; 5, ajmalicine; 6, speciociliatine; 7, paynantheine; 8, speciogynine, 9, mitragynine and 10, corynantheidine.

[0069] FIG. **2** is a scheme depicting a biosynthetic pathway of monoterpene alkaloids in *Mitragyna speciosa*. The inset shows an example of oxindole alkaloids from kratom which are synthesized via a separate pathway.

[0070] FIG. **3**A is a scheme depicting the synthesis of strictosidine by MsSTR-1 and the reaction catalyzed by the identified sequence. FIG. **3**B is a graph showing overlapped HPLC chromatograms showing the reactions with recombinant enzyme (MsSTR-1/c133034 without boiling (light line profile showing strictosidine) or after boiling the enzyme (darker line profile with the peaks of the substrates secologanin and tryptamine). FIG. **3**C. LC includes mass spectra obtained with the standard and the peak collected from the HPLC, confirming the identity of the product as strictosidine.

[0071] FIG. **4**A is a scheme depicting the conversion reaction of stictosidine aglycone to 19Z-geissoschizine catalyzed by the identified sequence. FIG. **4**B includes graphs of HPLC chromatograms showing the products of the reactions obtained with different amounts of recombinant enzyme (MsGSS-1/c101879) and in different reaction times. FIG. **4**C is a graph of the calculated area for the HPLC peaks for the products which co-elute with the standard.

[0072] FIG. **5**A is a graph of the reaction assay quantified with the HPLC method as described in FIG. **4** showing overlapped HPLC chromatograms showing the reactions with recombinant enzyme (MsGSS-1/c101879 without boiling (line profile with the peaks indicated by the arrows) or after boiling the enzyme as control (base line). The three peaks indicated by arrows were collected for identification by LC-MS. FIG. **5**B is a graph of the mass spectra profile obtained with the peak eluted at 12.9 min from the HPLC, confirming the identity of the product as 19Z-geissoschizine.

DETAILED DESCRIPTION

[0073] The diverse biological activities of monoterpene indole alkaloids have prompted the utilization of plants containing these as medicinal plants and as ingredients in products for treating pain, or addiction problems. Some show promise as important compounds for the development of new pharmacological agents or nutraceuticals for the treatment of different medical conditions. However, several obstacles have limited the potential applications of these compounds:

[0074] Apart from a small number of alkaloids that are relatively abundant in kratom leaves, most of these monoterpene alkaloids often exist at trace levels or are not present in all kratom varieties.

[0075] Extraction of these scarce but valuable alkaloids from the plant poses a significant challenge and their purification from mixtures is also complicated and costly.

[0076] Isolation from the kratom plant implies access to biomass which may not available due to legal or operational concerns.

[0077] In many cases, organic chemical synthesis is not an amenable cost-effective approach.

[0078] Described herein are enzymes that can serve as production catalysts for the in vitro and in vivo modification of indole alkaloid precursors such as strictosidine to produce unique indole alkaloids present in the plant *M. speciosa*. In addition, the polypeptide sequences described herein can be used in reactions for modifying similar molecules and generate novel synthetic indole alkaloid derivatives presenting similar, attenuated, or enhanced analgesic activities.

[0079] Thus, described herein are polypeptides that can act as enzymes, and related methods for the synthesis of monoterpene indole alkaloids. These polypeptides are typically derived from plant sources, such as *Mitragyna speciosa*. The use of enzymes with potential strictosidine reductase, strictosidine glucosidase, geissoschizine synthase, hydroxylase, and O-methyltransferase activity are described herein, for example, MsSTR1, MsSGD, MsGSS1, homologs of T16H, T16OMT, LAOMT, and related sequences.

[0080] In vitro catalysis systems, designed to utilize these enzymes and thereby produce substantially pure monoterpene indole alkaloids, such as strictosidine, strictosidine aglycone, 4,21-dehydrogeissoschizine, geissoschizine, corynantheidine, 9-hydroxycorynantheidine, mitragynine, 7-hydroxymitragynine, speciociliatine, speciogynine, and paynantheine, are described.

[0081] Like mitragynine, these and other monoterpene indole alkaloids may find use in compositions and methods to treat pain and to reduce dependence on opioids and opiates.

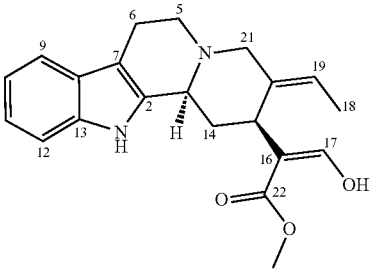
Definitions

[0082] Unless defined otherwise, technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. See, e.g. Singleton et al., Dictionary of Microbiology and Molecular Biology 2nd ed., J. Wiley & Sons (New York, N.Y. 1994); Sambrook et al., Molecular Cloning. A Laboratory Manual, Cold Springs Harbor Press (Cold Springs Harbor, NY 1989), each of which is incorporated herein by reference. For the purposes of the present application, the following terms are defined below.

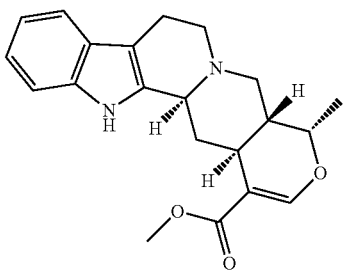
[0083] As used herein, the term "monoterpene indole alkaloid" refers to compounds containing a bicyclic structure of a benzene ring fused to a five-membered pyrrole ring. These compounds may be of natural or synthetic origin. In some examples, such compounds have a chemical structure derived from strictosidine. Some examples of monoterpene indole alkaloids include those typically found in *M. speciosa*, such as geissoschizine, ajmalicine, yohimbine, reserpine, sarpagin, mitragynine, cathenamine, catharanthine, tabersonine, vindolin, vincamine. In a particular aspect, these embodiments refer to the corynanthe-type of monoterpenoid indol alkaloids, for example, strictosidine, strictosidine aglycone, geissoschizine, corynantheidine, 9-hydroxycorynantheidine, mitragynine, 7-hydroxymitragynine, speciociliatine, speciogynine, and paynantheine.

[0084] The following structures of Geissoschizine and Ajmalicine are examples of monoterpene indole alkaloids. The numbering in the ring structure of Geissoschizine indicates positions where potential modifications such as hydroxylations and subsequent methylations can occur. As studied by Takayama (Takayama, 2004), the position of such modifications can affect the bioactivity of the molecule.

(A) Geissoschizine



(B) Ajmalicine



Examples of Corynanthe-Type Monoterpenoid Indole Alkaloids

[0085] As used herein, the term "variants" refers to variants of the sequences described herein. The term "variants" refers to biologically active sequences that have a peptide sequence that differs from the sequence of a native or wild-type sequence, by virtue of an insertion, deletion, modification, and/or substitution of one or more amino acids within the native sequence. Such variants have less than 100% sequence identity with a native sequence. Ordinarily, however, a biologically active variant will have an amino acid sequence with at least about 70% sequence identity with the sequence of a corresponding naturally occurring sequence, typically at least about 75%, more typically at least about 80%, even more typically at least about 85%, even more typically at least about 90%, and even more typically of at least about 95%, 96%, 97%, 98%, or 99% sequence identity. The polypeptide variants can include fragments of any length that retain the biological activity of the corresponding native sequence. Variants can also include sequences wherein one or more amino acids are added at either end of, or within, a native sequence. Variants can also include sequences where a number of amino acids are deleted and optionally substituted by one or more different amino acids.

[0086] As used herein, the term "treatment" or "therapy" refers to an approach for obtaining beneficial or desired clinical results. For the purposes described herein, beneficial or desired clinical results include, but are not limited to, alleviation of symptoms, diminishment of the extent of disease, stabilized (i.e., not worsening) state of disease, delay or slowing of disease progression, amelioration or palliation of the disease state, and remission (whether partial or total), whether detectable or undetectable. "Treatment" and "therapy" can also mean prolonging survival as compared to expected survival if not receiving treatment or therapy. Thus, "treatment" or "therapy" is an intervention performed to alter the pathology of a disorder. Specifically, the treatment or therapy may directly prevent, slow down or otherwise decrease a disorder such as pain, or may render the pain more susceptible to treatment or therapy by other therapeutic agents.

[0087] The terms "therapeutically effective amount", "effective amount" or "sufficient amount" mean a quantity sufficient, when administered to a subject, including a mammal, for example, a human, to achieve the beneficial or desired result, for example, an amount effective to treat pain. Effective amounts of the alkaloid compounds described herein may vary according to factors such as the disease state, age, sex, and weight of the subject. Dosage or treatment regimens may be adjusted to provide the optimum therapeutic response, as is understood by a skilled person.

[0088] Likewise, an "effective amount" of the indole alkaloid compounds described herein refers to an amount sufficient to function as desired, such as to treat pain.

[0089] Administration "in combination with" one or more further therapeutic agents includes simultaneous (concurrent) and consecutive administration in any order.

[0090] The term "pharmaceutically acceptable" means that the compound or combination of compounds is compatible with the remaining ingredients of a formulation for pharmaceutical use and that it is generally safe for administering to humans according to established governmental standards, including those promulgated by the United States Food and Drug Administration.

[0091] "Carriers" as used herein include cosmetically or pharmaceutically acceptable carriers, excipients, or stabilizers that are non-toxic to the cell or subject being exposed

thereto at the dosages and concentrations employed. Often the pharmaceutically acceptable carrier is an aqueous pH buffered solution. Examples of pharmacologically acceptable carriers include buffers such as phosphate, citrate, and other organic acids; antioxidants including ascorbic acid; low molecular weight (less than about 10 residues) polypeptides; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, arginine or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, and dextrins; chelating agents such as EDTA; sugar alcohols such as mannitol and sorbitol; salt-forming counterions such as sodium; and/or non-ionic surfactants such as TWEEN™, polyethylene glycol (PEG), and PLURONICS™.

[0092] In understanding the scope of the present application, the articles "a", "an", "the", and "said" preceding an element are intended to mean that there are one or more of the elements. Additionally, the term "comprising" and its derivatives, as used herein, are intended to be open-ended terms that specify the presence of the stated features, elements, components, groups, integers, and/or steps, but do not exclude the presence of other unstated features, elements, components, groups, integers and/or steps. The foregoing also applies to words having similar meanings such as the terms, "including", "having", and their derivatives.

[0093] It will be understood that any embodiments described as "comprising" certain components may also "consist of" or "consist essentially of," whereas "consisting of" has a closed-ended or restrictive meaning and "consisting essentially of" means including the components specified but excluding other components except for materials present as impurities, unavoidable materials present as a result of processes used to provide the components, and components added for a purpose other than achieving the technical effect of the invention. For example, a composition defined using the phrase "consisting essentially of" encompasses any known pharmaceutically acceptable additive, excipient, diluent, carrier, and the like. In some embodiments, a composition consisting essentially of a set of components will comprise less than 5% by weight, typically less than 3% by weight, more typically less than 1% by weight of non-specified components.

[0094] In addition, all ranges given herein include the end of the ranges and also any intermediate-range points, whether explicitly stated or not.

[0095] Finally, terms of degree such as "substantially", "about" and "approximately" as used herein mean a reasonable amount of deviation of the modified term such that the end result is not significantly changed. These terms of degree should be construed as including a deviation of at least ±5% of the modified term if this deviation would not negate the meaning of the word it modifies.

Polypeptides

[0096] Described herein, in aspects, are polypeptides encoding enzymes. These polypeptides are typically derived from plants such as *Mitragyna speciosa* and find use in producing monoterpene indole alkaloids with medicinal activity and/or precursors to such compounds. For example, described herein are polypeptides comprising one or more of the following sequences, variants thereof, or fragments of the polypeptides or variants. Variants can be of natural origin such as orthologous sequences from related plant species, or of non-natural origin obtained synthetically or by mutations. The following sequences, are shown in attached Sequence Listing.

[0097] SEQ IDs No: 1-13—polypeptide sequences orthologous to strictosidine-O-beta-D-glucosidase (SGD)

[0098] SEQ IDs No: 14-22—polypeptide sequences orthologous to geissoschizine synthase (GSS/THAS)

[0099] SEQ IDs No: 23-54—polypeptide sequences homologous to tabersonine 16-hydroxylase (T16H)

[0100] SEQ IDs No: 55-61—polypeptide sequences homologous to tabersonine 16-O-methyltransferase (T16OMT)

[0101] SEQ IDs No: 62-65—polypeptide sequences homologous to loganic acid O-methyltransferase (LAOMT)

[0102] The variants of the polypeptides described herein may have any degree of sequence identity to the polypeptides, provided they retain some degree of native activity of the corresponding native sequence, for example strictosidine reductase, strictosidine glucosidase, geissoschizine synthase, hydroxylase, or O-methyltransferase activity. For example, the variants typically have at least about 70%, about 71%, about 72%, about 73%, about 74%, about 75%, about 76%, about 77%, about 78%, about 79%, about 80%, about 81%, about 82%, about 83%, about 84%, about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or about 99.5% sequence identity.

[0103] Likewise, the fragments of the polypeptides or variants described herein may have any length, provided they retain some degree of native activity, for example, strictosidine reductase, strictosidine glucosidase, geissoschizine synthase, hydroxylase, or O-methyltransferase activity. For example, the fragments may be missing about 1, about 2, about 3, about 4, about 5, about 6, about 7, about 8, about 9, about 10, about 11, about 12, about 13, about 14, about 15, about 16, about 17, about 18, about 19, about 20, about 21, about 22, about 23, about 24, about 25, about 26, about 27, about 28, about 29, about 30, about 31, about 32, about 33, about 34, about 35, about 36, about 37, about 38, about 39, about 40, about 41, about 42, about 43, about 44, about 45, about 46, about 47, about 48, about 49, about 50, about 60, about 70, about 80, about 90, about 100, about 125, about 150, about 175, about 200, or about 250 amino acid residues as compared to the polypeptide in question.

[0104] The polypeptides, variants, and fragments described herein may also be fused to other polypeptides and could, therefore comprise additional amino acid residues, such as about 1, about 2, about 3, about 4, about 5, about 6, about 7, about 8, about 9, about 10, about 11, about 12, about 13, about 14, about 15, about 16, about 17, about 18, about 19, about 20, about 21, about 22, about 23, about 24, about 25, about 26, about 27, about 28, about 29, about 30, about 31, about 32, about 33, about 34, about 35, about 36, about 37, about 38, about 39, about 40, about 41, about 42, about 43, about 44, about 45, about 46, about 47, about 48, about 49, about 50, about 60, about 70, about 80, about 90, about 100, about 125, about 150, about 175, about 200, about 250, about 300, about 400, about 500, about 600, about 700, about 800, about 900, or about 1000 or more additional amino acids.

[0105] The polypeptides described herein may comprise, consist of, or consist essentially of the sequences of SEQ ID NO: 1-65, and typically encode an enzyme.

[0106] The polypeptides described herein are typically expressed in a host cell or organism, such as a bacterium, a yeast, an alga, a fungus, or a plant, including single cells and cell cultures of any thereof for enzymatically acting on a molecule present in the host cell or organism or its cell culture medium. The polypeptides described herein may instead be used in a cell-free system for acting on a molecule present in the system.

[0107] In embodiments, the hosts described herein endogenously express and/or are engineered to express at least one nucleic acid coding for an enzyme with strictosidine reductase, strictosidine glucosidase, geissoschizine synthase, hydroxylase, or O-methyltransferase activity or variants thereof to produce a monoterpene indole alkaloid, such as strictosidine, strictosidine aglycone, geissoschizine, corynantheidine, 9-hydroxycorynantheidine, mitragynine, 7-hydroxymitragynine, speciociliatine, speciogynine, or paynantheine, or variants thereof.

[0108] Host cells described herein can be any cell capable of producing at least one protein described herein and include bacterial, fungal (including yeast), animal, algal, and plant cells. The cells may be prokaryotic or eukaryotic. Typical host cells are bacterial, yeast, algal, and plant cells. In a typical embodiment, the plant cell is a seed cell, in particular, a cell in a cotyledon or endosperm of a seed. In one embodiment, the cell is a bacterial cell. An example of a bacterial cell useful as a host cell of the present invention is *Escherichia coli*, Synechococcus spp. (also known as *Synechocystis* spp.), for example, Synechococcus *elongatus*. Examples of algal cells useful as host cells of the present invention include, for example, *Chlorella* sp., *Chlamydomonas* sp. (for example, *Chlamydomonas reinhardtii*), *Dunaliella* sp., and Haematococcus sp. It will be understood that some host cells may not be able to produce an alkaloid because they lack the necessary precursors but can be used as host cells to produce the polypeptides described herein. Thus, it will be understood that the polypeptides described herein can be expressed in a variety of expression host cells e.g., bacteria, yeasts, plant cells, algal cells, or cell-free expression systems. In one embodiment, described herein are expression vectors comprising the coding DNA sequence for the polypeptides described herein for the expression and purification of the recombinant polypeptide produced from a protein expression system using host cells selected from, e.g., bacteria, mammalian, insect, yeast, or plant cells. In some embodiments, the nucleic acid can be sub-cloned into a recombinant expression vector that is appropriate for the expression of fusion polypeptide in bacteria, mammalian, yeast, or plant cells or a cell-free expression system such as the wheat germ cell-free expression system or a rabbit reticulocyte expression system.

[0109] In some embodiments, cell-free systems can include in vitro enzymatic reactions performed in tubes, columns, chips, or any other solid support or surface where a polypeptide described herein is present in solution or immobilized in a resin or another solid support matrix. A range of reversible physical adsorption and ionic linkages, to irreversible stable covalent bonds exist to produce immobilized enzymes.

[0110] Thus, in embodiments, the host cells or cell-free systems described herein are suitable for producing a substantially pure polypeptide and/or substantially pure indole alkaloid in the presence of at least an enzyme such as those represented by SEQ ID: 1-75 (or fragments or variants thereof).

[0111] In embodiments, the cell-free systems can include a previous modification step such as a nucleic acid coding for secologanin synthase to produce secologanine and for tryptophan decarboxylase to produce tryptamine which can then both be used by a polypeptide described herein to synthesize a substantially pure strictosidine.

[0112] In embodiments, the production of indole alkaloid molecules with the polypeptides described herein are carried out following a unique order of reactions, but such polypeptides can be used with different substrates in the described pathway or similar molecules to produce the described indole alkaloids or variants.

[0113] In embodiments, the enzymatic reactions catalyzed by these polypeptides can also be combined employing organic chemistry synthesis methods.

[0114] Also described herein are compositions comprising an indole alkaloid obtainable or obtained by one of the methods as disclosed above, and the use of the said composition as a medicinal agent, such as an analgesic agent, for pharmacological purposes.

[0115] Provided herein are compositions comprising substantially pure indole alkaloids, which are, for example, at least about 95%, about 96%, about 97%, about 98%, about 99%, about 99.5%, or about 99.9% pure.

[0116] The compositions comprising an indole alkaloid described herein may be formulated for use by a subject, such as a mammal, including a human. Such compositions may comprise about 0.00001% to about 99% by weight of the active compound(s) and any range there-in-between. For example, typical doses may comprise from about 0.1 μg to about 100 μg of the molecules described herein per 300 mg dose, such as about 0.5 μg, about 1 μg, about 2 μg, about 3 μg, about 4 μg, about 5 μg, about 6 μg, about 7 μg, about 8 μg, about 9 μg, about 10 μg, about 25 μg, about 50 μg, or about 75 μg per 300 mg dose, such as from about 0.1 μg to about 10 μg, or from about 1 μg to about 5 μg, or from about 1 μg to about 2 μg per 300 mg dose (and all related increments and percentages by weight).

[0117] The alkaloid molecules described herein may be used in any suitable amount, but are typically provided in doses comprising from about 1 to about 10000 ng/kg, such as from about 1 to about 1000, about 1 to about 500, about 10 to about 250, or about 50 to about 100 ng/kg, such as about 1, about 10, about 25, about 50, about 75, about 100, about 150, about 200, about 250, about 300, or about 500 ng/kg. Similar amounts, higher amounts, or lower amounts could be used for administration.

[0118] The indole alkaloid molecules described herein may be administered over a period of hours, days, weeks, or months, depending on several factors, including the severity and type of pain or other condition being treated, whether a recurrence is considered likely, or to prevent pain or other condition, etc. The administration may be constant, e.g., constant infusion over a period of hours, days, weeks, months, etc. Alternatively, the administration may be intermittent, e.g., the molecules may be administered once a day over a period of days, once an hour over a period of hours, or any other such schedule as deemed suitable.

[0119] The compositions described herein can be prepared by per se known methods for the preparation of pharma-

ceutically or cosmetically acceptable compositions which can be administered to subjects, such that an effective quantity of the active substance is combined in a mixture with a pharmaceutically acceptable vehicle. Suitable vehicles are described, for example, in "Handbook of Pharmaceutical Additives" (compiled by Michael and Irene Ash, Gower Publishing Limited, Aldershot, England (1995)). On this basis, the compositions include, albeit not exclusively, solutions of the substances in association with one or more pharmaceutically acceptable vehicles or diluents, and may be contained in buffered solutions with a suitable pH and/or be iso-osmotic with physiological fluids.

[0120] Pharmaceutically acceptable carriers are well known to those skilled in the art and include, for example, sterile saline, lactose, sucrose, calcium phosphate, gelatin, dextrin, agar, pectin, peanut oil, olive oil, sesame oil, *cannabis* oil, and water. Furthermore, the composition may comprise one or more stabilizers such as, for example, carbohydrates including sorbitol, mannitol, starch, sucrose, dextrin, and glucose, proteins such as albumin or casein, and buffers like alkaline phosphates.

[0121] The indole alkaloid molecules described herein can, in embodiments, be administered for example, by parenteral, intravenous, subcutaneous, intradermal, intramuscular, intracranial, intraorbital, ophthalmic, intraventricular, intracapsular, intraspinal, intracisternal, intraperitoneal, intranasal, intrarectal, intravaginal, aerosol, oral, topical, or transdermal administration. Typically, the compositions of the invention are administered orally.

[0122] It is understood by one of skill in the art that the produced molecules described herein can be used in conjunction with known therapies for prevention and/or treatment of pain in subjects. Similarly, the produced modified molecules described herein can be combined with one or more other pharmaceutical or natural health products, such as cannabinoids, terpenes, or other natural or synthetic compounds. The produced molecules described herein may, in embodiments, be administered in combination, concurrently or sequentially, with conventional treatments for pain, including non-steroidal anti-inflammatory drugs, for example. The indole alkaloid molecules described herein may be formulated together with such conventional treatments when appropriate. Other uses of these alkaloid molecules can be found due to their anticipated antidepressant, anxiolytic, antipsychotic, antihypertensive, and antipyretic activities, or for the treatment or withdrawal from opioid and alcohol dependence and abuse.

[0123] The above disclosure generally describes the present invention. A more complete understanding can be obtained by reference to the following specific Examples. These Examples are described solely for purposes of illustration and are not intended to limit the scope of the invention. Changes in form and substitution of equivalents are contemplated as circumstances may suggest or render expedient. Although specific terms have been employed herein, such terms are intended in a descriptive sense and not for purposes of limitation.

Methods

Plant Material and Growth Conditions.

[0124] *Mitragyna speciosa* (Korth.) Havil plants (Malay and Rifat varieties) can be maintained in growth chambers with a 16-hour light photoperiod (200 μmol $m^{-2}s^{-1}$; mixed cool white and incandescent bulbs), a day/night temperature regime of 28° C./26° C., and a constant relative humidity of 80%. Plants are grown in a 2:1:1 (v/v) mixture of coco coir (Millennium soils Coir): perlite (Therm-o-rock East Inc.): turface (Turface Athletics) before harvesting material. The plant material was identified and authenticated by Dr. Carole Ann Lacroix and a voucher specimen (No. 102033) was deposited at the Ontario Agricultural College Herbarium in Guelph, Ontario, Canada.

Mitragyna Transcript Sequences.

[0125] In typical aspects, total mRNA is extracted from young and mature leaves using the RNeasy Plant mRNA extraction kit (Qiagen). Quality control, normalization, mRNA Seq library construction, and de novo transcriptome assembly were performed by Macrogen Inc. (macrogen.com; Seoul, Korea) in order to reconstruct transcript sequences without a reference genome. Transcripts are represented by assembled contigs and then merged using the Trinity program, and then clustered into non-redundant transcripts or unigenes using CD-HIT-EST program with a minimum length of 200 bp. The obtained unigenes can then be used for the subsequent annotation, ORF (open reading frame) prediction, and expression analysis. ORF prediction for the unigenes is performed using TransDecoder program to identify candidate coding regions.

Identification of Enzymes for Indole Alkaloid Biosynthesis in *Mitragyna*.

[0126] Nucleotide sequences encoding for putative enzymes required for the different steps for the biosynthesis of mitragynine and related alkaloids can be obtained by searching for homologous sequences in the de novo transcriptome analysis described above using sequences of other plant species known to participate in alkaloid metabolism. The tblastn queries were perform with *Catharanthus roseus*' strictosidine-O-beta-D-glucosidase (B8PRP4), geissoschizine synthase (crGS1splW8JWW7), tabersonine 16-hydroxylase 1 (crT16H1splP98183), tabersonine 16-hydroxylase 2 (crT16H2splU5HKE8), tabersonine 16-O-methyltransferase (cr16OMTsplBOEXJ8), and loganic acid O-methyltransferase (crLAMTsplB2KPR3); strictosidine synthase (rvSTR_spIP68174) from Rauvolfia *mannii*, and strictosidine-O-beta-D-glucosidase (rsSGD_splQ8GU20) from Rauvolfia *serpentina*. In addition, other orthologous sequences were obtained with tryptophan decarboxylase (msTDC_trIG5DF84) and secologanin synthase (msSLS3_trlH9MZK1) from *Mitragyna speciosa*. When possible, the sequences were then matched against the *Mitragyna* genome contig database, via BLASTn searches, to retrieve the full-length open reading frames for the selected sequences. The in-silico assembled amino acid sequences from these gene sequences were then used to construct phylogenetic relationships using the MEGA software package (version 6.0) by the neighbor-joining method with bootstrap analysis of 1000 replicates.

Expression of Recombinant in *E. coli*.

[0127] In typical aspects, open reading frames encoding the selected candidate enzymes are synthesized commercially. Each cDNA is amplified by PCR and then inserted into an expression vector system (for example, Novagen's pET28) which introduces an N-terminal 6×His tag to the coding sequence. The construct is then introduced into a

bacterial host (for example *E. coli* BL21-CodonPlus (DE3)-RIPL cells). Bacterial cells expressing a recombinant enzyme are cultured in LB media for several hours in the presence of IPTG to induce recombinant protein expression. The bacterial cells are collected by centrifugation, re-suspended, and then disrupted by sonication. Crude protein extracts are applied and purified with a Ni^{2+} affinity matrix (for example a HisTrap FF column). Afterward, the enzyme is eluted with high concentrations (250 to 400 uM) of imidazole, and then immediately desalted on PD-10 columns equilibrated in a suitable buffer. Protein concentration is determined by Bradford Assay using BSA as a standard. The purified protein can then be frozen before use.

Chemicals and Reagents.

[0128] Authentic alkaloid standards can be purchased from specialized chemical companies: Ajmalicine (Sigma-Aldrich) and mitragynine, 7-hydroxymitragynine, paynantheine, speciogynine, mitraphylline, speciociliatine, tryptamine, secologanin, strictosidine-aglycone, and 19Z-geissoschizine (Cayman Chemicals). Other chemicals used in enzymatic reactions can be procured from Sigma-Aldrich. These external alkaloid standards can be used for quantification based on the peak area revealed by the HPLC analyses.

Mass Spectrometry Analysis of Enzymatic Reaction Products.

[0129] Generally, samples are evaporated under nitrogen and then re-suspended in methanol prior to liquid chromatography-mass spectrometry (LC-MS) analysis (for example HPLC (high performance liquid chromatography) interfaced with a Q-TOF (time-of-flight) mass spectrometer). Typically, samples can be resolved with a reverse-phase column and eluted over a gradient from 45% to 95% methanol with 0.1% formic acid (v/v), or with 1:1 water and acetonitrile as solvents, both with 0.1% formic acid, followed by 100% methanol for 10-20 min. The eluted products are detected by absorption at the 210-350 nm range and quantified relative to authentic standards based on the peak area of the HPLC analysis. During the LC-MS analysis, the mass-to-charge ratio is typically scanned across the m/z range of 100-3000 m/z in an extended dynamic range positive-ion MS mode. Chromatograms can be analyzed by using a software that compares MS patterns from standard libraries or known standards used in the laboratory. Fragmentation patterns of the various parent (molecular) ions obtained using collision energies of 5 to 20 V from the recovered peak products (modified compounds) are usually also compared with fragmentation patterns of standard molecules.

[0130] In typical aspects, nuclear magnetic resonance (NMR) spectroscopy, is used as a preeminent technique for determining the structure of the indole alkaloids that can be obtained (for example for determining the position of double bonds, hydroxyl, or methyl groups). After the enzymatic reaction products from the enzymatic assays are resolved by HPLC, the compounds are eluted and subsequently collected. Usually, approximately 0.5 mg of each compound is evaporated to dryness under N_2 gas, resuspended in acetone-d6, and analyzed using ^{1}H and ^{13}C NMR. NMR spectra are collected on a spectrometer (for example a Bruker AVANCE III 600 MHz equipped with a 5 mm TCI cryoprobe).

[0131] This or a similar in vitro cell-free system described herein can be suitable for producing a substantially pure indole alkaloid in the presence of at least an enzyme described herein.

EXAMPLES

Example 1

[0132] This example describes a general method for analyzing alkaloids from *Mitragyna speciosa*, which also validates the process used for the identification of enzymatic products generated below. About 100 g of dried leaf material was extracted with 2 L of an acetic acid solution (0.5 M) at 80° C. for 30 min. The crude extract was filtered and then passed through a 60 ml column containing polyvinylpyrrolidone (PVPP, 110 μm particle size) to remove polyphenolic compounds and then sequentially chromatographed over 50 ml of Diaion HP-20 resin (Supelco) equilibrated with distilled water and the reversed-phase column was then washed with 20% (v/v) methanol before elution with 100% methanol followed by methanol/ethyl acetate (50:50 v/v). The recovered alkaloid fractions were pooled and reduced to 200 ml on a rotary evaporator at 70° C. The isolate was then loaded onto an ion-exchange resin (AmberChrom 50WX2, 200-400) and washed with 500 ml of acetic acid in ethanol (0.025 M), followed by 250 ml of 100% ethanol. Alkaloids were cluted with 340 ml of 2.8 M ammonium hydroxide in ethanol and then brought to a 150 ml final volume. This purified alkaloid extract was subjected to phase separation with chloroform (300 ml). The organic layer was extracted and reduced to dryness, in vacuo, and resuspended in 10 ml of 0.2 M HCl. After complete resuspension, the pH was adjusted to 5.0. Alkaloids fractions were analyzed using an Agilent 1260 Infinity liquid chromatography system equipped with a reversed-phase Kinetex EVO C18 100 Å column (150×4.6 mm, 5 μm). Separation of the alkaloids was achieved using a binary gradient with ammonium bicarbonate buffer (5 mM pH 9.5; A) and acetonitrile (B), starting with 70% solvent A transitioning to 70% solvent B for 18 min at a flow rate of 1.5 ml/min. Alkaloids were quantified at 226 nm. Chemical structures of common alkaloids detected in kratom are shown in FIG. **1**A and an example of an HPLC chromatogram illustrating the alkaloid profile obtained with a purified crude extract is presented in FIG. **1**B, where different peaks (top panel) can be compared with standards (bottom panel). For example, Mitragynine clutes at 14.4 min.

[0133] To verify the identity of the molecules, the alkaloid fractions were collected and approximately 0.5 mg of each compound were evaporated to dryness, resuspended in deuterated chloroform, and analyzed using ^{1}H NMR. NMR spectra were collected on a Bruker AVANCE III 600 MHz spectrometer equipped with a 5 mm TCI cryoprobe. According to their chemical structure, and to what has been described in *Catharanthus roseus* for the synthesis of alkaloids (e.g. cathenamine), a biosynthetic pathway of monoterpene indole alkaloids in *M. speciosa* is proposed (FIG. **2**). From it, enzyme-encoding homologous sequences were searched in the transcriptome and potential candidates were identified (Appendix A).

[0134] The following examples describe general methods for using polypeptide sequences from *Mitragyna speciosa* to produce monoterpene indole alkaloids and its intermediaries.

Example 2

[0135] This example describes a general method for using strictosidine reductase polypeptide sequences from *Mitragyna speciosa* to produce strictosidine from secologanine and tryptamine (FIG. **3**A). A synthetic cDNA sequence of the MsSTR-1 sequence was sub-cloned in the pET28a vector system (Novagen) which introduced an N-terminal 6×His tag to the coding sequence and then the construct was introduced into *E. coli* BL21-CodonPlus (DE3)-RIPL cells. Bacterial cells expressing recombinant MsSTR-1 were used to produce recombinant MsSTR-1 protein as described in the Methods above, and stored at −80° C. before use.

[0136] The enzymatic assays with recombinant enzyme MsSTR-1 consisted of a total volume of 100 μL of 2 mM of secologanin, 1 mM tryptamine, and 10 μg of recombinant STR enzyme in 100 mM NaH_2PO_4 buffer pH 6.8 at 30° C. for 24 h. A control reaction was performed with boiled enzyme. Reactions were terminated by boiling for 2 min, centrifuged for 5 min and 20 μL was injected on an HPLC. Strictosidine was analyzed using an Agilent 1260 Infinity liquid chromatography system equipped with a reverse phase Waters Spherisorb 5 μM (ODS2 4.6×250 mm) column. The chromatographic separation of tryptamine, secologanin and strictosidine was achieved using a binary gradient with (0.1% TFA; A) and (acetonitrile; B) gradient at a flow rate of 1 mL/min. The amount of solvent A in the mobile phase under the starting conditions was 68% and decreased to 30% until 16 min. Strictosidine and tryptamine were detected by absorption at 280 nm, and the amount of strictosidine produced was reported in equivalents of tryptamine. The HPLC chromatogram of the product showed a major peak corresponding to stroctisidine according to the eluted time of the standard (FIG. **3**B). The peak was collected and processed for mass spectrometry analysis.

[0137] Samples were evaporated under nitrogen and then re-suspended in methanol prior to liquid chromatography-mass spectrometry (LC-MS) analysis performed on an Agilent 1200 HPLC interfaced with an Agilent UHD 6530 Q-TOF mass spectrometer. A C18 cartridge column (Agilent Rapid Resolution 2.1×30 mm, 3.5 μm) was used at 30° C. with 1:1 water and acetonitrile as solvents, both with 0.1% formic acid. The flow rate was maintained at 0.4 ml/min. The mass spectrometer electrospray capillary voltage was maintained at 4.0 Kv and the drying gas temperature at 250° C. with a flow rate of 8 L/min. Nitrogen was used as both nebulizing, drying gas, and collision-induced dissociation gas. The instrument was externally calibrated with the ESI TuneMix (Agilent). Chromatograms were analyzed within Agilent Qualitative Analysis software B 08.0. Fragmentation patterns of the various parent (molecular) ions were obtained using collision energies of 5, 10 and 20 V. HPLC profile and the mass spectral fragmentation pattern obtained from the recovered peak product of the assay (FIG. **3**C) corresponded to the mass spectra observed with the strictosidine standard.

Example 3

[0138] This example describes a general method for using geissoschizine synthase polypeptide sequences from *Mitragyna speciosa* to produce geissoschizine from strictosidine aglycone (FIG. **4**A). A synthetic cDNA sequence of the MsGSS-1 sequence was sub-cloned in the pET28a vector system (Novagen) which introduced an N-terminal 6×His tag to the coding sequence and then the construct was introduced into *E. coli* BL21-CodonPlus (DE3)-RIPL cells. Bacterial cells expressing recombinant MsGSS-1 were used to produce a recombinant MsGSS-1 protein as described in the Methods above, and stored at −80° C. before use.

[0139] Enzymatic reactions consisted in a total volume of 100 μL of 135 μM of strictosidine-aglycone, 200 μM of NADPH and 2-8 μg of recombinant MsGSS-1 enzyme in 20 mM Tris-HCl buffer pH 7.5 at 30° C. for 1 h and 6 h. A control reaction was performed with boiled enzyme. Reactions were terminated by boiling for 2 min, centrifuged for 5 min and extracted twice with ethyl-acetate. After drying under nitrogen gas, samples were resuspended in 60 μl methanol and 20 μl were injected in the HPLC. Geissoschizine was analyzed using an Agilent 1260 Infinity liquid chromatography system equipped with a reverse phase column Kinetex EVO C18 100 Å (150×4.6 mm, 5 μm). Chromatographic separation of geissoschizine was achieved using a binary gradient with ammonium bicarbonate buffer (5 mM, pH 9.5; A) and acetonitrile (B), starting with 70% solvent A transitioning to 70% solvent B over the course of 17 minutes at a flow rate of 1.5 mL/min. 19Z-geissoschizine was quantified at 226 nm. The chromatograms of the products showed the same peak (FIG. **4**B), and peak areas of all five reactions (control, 2 μg protein 1 h, 2 μg protein 6 h, 8 μg protein 1 h, and 8 μg protein 6 h) were calculated (FIG. **4**C). Representative peaks were collected and processed for mass spectrometry analysis. For this, samples were evaporated under nitrogen and then re-suspended in methanol before liquid chromatography-mass spectrometry (LC-MS) analysis was performed on an Agilent 1200 HPLC interfaced with an Agilent UHD 6530 Q-TOF mass spectrometer, using the conditions described in the previous example. The chromatogram of the reaction product showed three peaks that eluted at 1.8, 2.5, and 12.9 min in samples that had the 'live' enzyme and not in samples with boiled enzyme (FIG. **5**A). HPLC profile and the mass spectral fragmentation pattern obtained from the recovered peak product at 12.9 min (FIG. **5**B) corresponded to the mass spectra of a geissoschizine standard.

[0140] The above disclosure generally describes the present invention. Although specific terms have been employed herein, such terms are intended in a descriptive sense and not for purposes of limitation.

[0141] All publications, patents, and patent applications cited herein are incorporated by reference in their entirety to the same extent as if each publication, patent, or patent application was specifically and individually indicated to be incorporated by reference in its entirety.

[0142] Although preferred embodiments of the invention have been described herein in detail, it will be understood by those skilled in the art that variations may be made thereto without departing from the spirit of the invention or the scope of the appended claims.

[0143] A sequence listing is being filed herewith, the sequence listing is incorporated herein by reference in its entirety.

REFERENCES

[0144] Adkins J E, Boyer E W, McCurdy C R. (2011) *Mitragyna speciosa*, a psychoactive tree from Southeast Asia with opioid activity. Curr Top Med Chem. 11:1165-1175.

[0145] Avery B A, Boddu S P, Sharma A, Furr E B, Leon F, Cutler S J, et al. (2019) Comparative pharmacokinetics of mitragynine after oral administration of *Mitragyna speciosa* (kratom) leaf extracts in rats. Planta Med. 2019 85:340-346.

[0146] Center of Disease Control. Opioid Crisis, Understanding the Epidemic. https://www.cdc.gov/opioids/basics/epidemic.html. Accessed Feb. 7, 2022

[0147] Coe M A, Pillitteri J L, Sembower M A, Gerlach K K, Henningfield J E (2019) Kratom as a substitute for opioids: Results from an online survey. Drug Alcohol Depend. 202:24-32.

[0148] Charoonratana T, Wungsintaweekul J, Keawpradub N, Verpoorte R (2013) Molecular cloning and expression of tryptophan decarboxylase from *Mitragyna speciosa*. Acta Physiol Plant 35:2611-2621.

[0149] De Luca V., Salim V., Thamm A., Masada S. A., Yu F. (2014) Making iridoids/secoiridoids and monoterpenoid indole alkaloids: Progress on pathway elucidation. Curr Opin Plant Biol. 19:35-42.

[0150] Feng L Y, Battulga A, Han E, Chung H, Li J H. (2017) New psychoactive substances of natural origin: A brief review. J. Food Drug Anal. 25: 461-471.

[0151] Field E. (1921) XCVIII.-Mitragynine and mitraversine, two new alkaloids from species of Mitragyne. J Chem Soc Trans. 119:887-891.

[0152] Flores-Bocanegra L, Raja H A, Graf T N, Augustinović M, Wallace E D, Hematian S, Kellogg J J, Todd D A, Cech N B, Oberlies N H (2020) The chemistry of kratom [*Mitragyna speciosa*]: updated characterization data and methods to elucidate indole and oxindole alkaloids. J Nat Prod. 83:2165-2177.

[0153] Gardner E A, McGrath S A, Dowling D, Bai D (2022) The Opioid Crisis: Prevalence and Markets of Opioids. Forensic Sci Rev. 34:43-70.

[0154] Griffin O H, Webb M E (2018) The scheduling of kratom and selective use of data. J Psychoactive Drugs 50:114-120.

[0155] Gutridge A M, Robins M T, Cassell R J, Uprety R, Mores K L, Ko M J, Pasternak G W, Majumdar S, van Rijn R M. (2020) G protein-biased kratom-alkaloids and synthetic carfentanil-amide opioids as potential treatments for alcohol use disorder. Br J Pharmacol. 177:1497-1513.

[0156] Hemby S E, McIntosh S, Leon F, Cutler S J, McCurdy C R (2019) Abuse liability and therapeutic potential of the *Mitragyna speciosa* (kratom) alkaloids mitragynine and 7-hydroxymitragynine. Addict Biol. 24:874-885.

[0157] Jumali S S, Said I M, Baharum S N, Ismail I, Rahman Z A, Zainal Z (2011) Molecular cloning and characterization of strictosidine synthase, a key gene in biosynthesis of mitragynine from *Mitragyna speciosa*. Afr J Biotech 10:15238-15244.

[0158] Kruegel A C, Gassaway M M, Kapoor A, Varadi A, Majumdar S, Filizola M, et al. (2016) Synthetic and receptor signaling explorations of the *Mitragyna* alkaloids: mitragynine as an atypical molecular framework for opioid receptor modulators. J Am Chem Soc. 138:6754-6764.

[0159] Kruegel A C, Grundmann O (2018) The medicinal chemistry and neuropharmacology of kratom: A preliminary discussion of a promising medicinal plant and analysis of its potential for abuse. Neuropharmacology 134 (A): 108-120.

[0160] Leon F, Habib E, Adkins J E, Furr E B, McCurdy C R, Cutler S J. (2009) Phytochemical characterization of the leaves of *Mitragyna speciosa* grown in U.S.A. Nat Prod Commun. 4:907-910.

[0161] Mafi A, Kim S K, Goddard W A 3rd. (2020) Mechanism of beta-arrestin recruitment by the mu-opioid G protein-coupled receptor. Proc Natl Acad Sci USA. 117:16346-16355.

[0162] Matsumoto K, Horie S, Ishikawa H, Takayama H, Aimi N, Ponglux D, et al. (2004) Antinociceptive effect of 7-hydroxymitragynine in mice: Discovery of an orally active opioid analgesic from the Thai medicinal herb *Mitragyna speciosa*. Life Sci. 74:2143-2155.

[0163] Meireles V, Rosado T, Barroso T, Soares S, Gonçalves J, Luís A, Caramelo D, Simão A Y, Fernández N, Duarte A P, Gallardo E (2019) *Mitragyna speciosa*: Clinical, Toxicological aspects and analysis in biological and non-biological samples. Medicines (Basel) 6:35.

[0164] Palamar J J (2021) Past-year kratom use in the U.S.: estimates from a nationally representative sample. Am J Prev Med. 61:240-245.

[0165] Prozialeck W C, Jivan J K, Andurkar S V (2012) Pharmacology of kratom: an emerging botanical agent with stimulant, analgesic and opioid-like effects. J Am Osteopath Assoc. 112:792-799.

[0166] Schotte C, Jiang Y, Grzech D, Dang T T, Laforest L C, León F, Mottinelli M, Nadakuduti S S, McCurdy C R, O'Connor S E. (2023) Directed biosynthesis of mitragynine stereoisomers. J Am Chem Soc. 145:4957-4963.

[0167] Sharma A, Kamble S H, León F, Chear N J, King T I, Berthold E C, Ramanathan S, McCurdy C R, Avery B A (2019) Simultaneous quantification of ten key Kratom alkaloids in *Mitragyna speciosa* leaf extracts and commercial products by ultra-performance liquid chromatography-tandem mass spectrometry. Drug Testing & Analysis 11:1162-1171.

[0168] Takayama H (2004) Chemistry and pharmacology of analgesic indole alkaloids from the rubiaceous plant, *Mitragyna speciosa*. Chem Pharm Bull. 52:916-928.

[0169] Treimer J F, Zenk M H (1979) Purification and properties of strictosidine synthase, the key enzyme in indole alkaloid formation. Eur J Biochem. 101:225-233.

[0170] Váradi A, Marrone G F, Palmer T C, Narayan A, Szabo M R, Le Rouzic V, et al. (2016) Mitragynine/corynantheidine pseudoindoxyls as opioid analgesics with mu agonism and delta antagonism, which do not recruit beta-arrestin-2. J Med Chem. 59:8381-8397.

[0171] Veeramohan R, Azizan K A, Aizat W M, Goh H H, Mansor S M, Yusof N S M, Baharum S N, Nga C L (2018) Metabolomics data of *Mitragyna speciosa* leaf using LC-ESI-TOF-MS. Data Brief 18:1212-1216.

[0172] Wilson L L, Chakraborty S, Eans S O, Cirino T J, Stacy H M, Simons C A, et al. (2021) Kratom alkaloids, natural and semi-synthetic, show less physical dependence and ameliorate opioid withdrawal. Cell Mol Neurobiol. 41:1131-1143.

[0173] World Health Organization. Opioid overdose, Key facts. https://www.who.int/news-room/fact-sheets/detail/opioid-overdose. Accessed Feb. 7, 2022

[0174] Yusoff N H, Suhaimi F W, Vadivelu R K, Hassan Z, Rumler A, Rotter A, et al. (2016) Abuse potential and adverse cognitive effects of mitragynine (kratom). Addict Biol. 21:98-110.

SEQUENCE LISTING

Sequence total quantity: 65

SEQ ID NO: 1 moltype = AA length = 509
FEATURE Location/Qualifiers
source 1..509
mol_type = protein
organism = Mitragyna speciosa
SEQUENCE: 1
MSTPPTKFSG TLSRSDFPEG FLFGSASSAF QYEGAHNVDG RLPSIWDTFL KETHPDIVAA 60
NGLDAVEFYY RYREDIKAMK DIGLDTFRFS LSWPRILPNG RRTKGPNNEE QGVNRLAIDF 120
YNTVINLLLE NGIEPSVTLF HWDLPQALET EYLGFLSQKS VEDFVDYADL CFREFGDRVK 180
YWMTFNETWS YSLFGYLLGT FAPGRGSTDE GQRRAMAEVL PSSLGRSKQA FAHSRTPRAG 240
DPTTEPYIVT HNQLLAHAAA TKLYRREYQN VQKGKMGIGL VTIWTEPHTD TQEDRDAAQR 300
VLDFMLGWLF DPVVFGKYPE SMRRLVGNRL PEFTPEQSRD LIGSFDFIGM NYYTANSVAN 360
RPYSRSIIYN PDSQAICFSL GEEAGSSWVY IYPQGLLKLL LYVKEKYNNP VIYITENGID 420
EVNDENLTMW EALYDTQRIS YHKQHIEATK QAISQGADVR GYYAWSFTDN LEWASGFDSR 480
FGLNYVHFGR ELERYPKLSA GWFKFFLDK 509

SEQ ID NO: 2 moltype = AA length = 544
FEATURE Location/Qualifiers
source 1..544
mol_type = protein
organism = Mitragyna speciosa
SEQUENCE: 2
MAASNVLSPI SRADFGEEFI FGSASSAYQM EGAAAEGGRG PSIWDTFTEQ TPDKVMDGSN 60
GNVAINQYYL YKEDVKLMKK IGLDAYRFSI SWPRVLPGGR LSAGVNKAGI DYYNNLINEL 120
LANDIKPFVT LFHWDVPQAL EDEYGGFLNR KIVNDFREFA ELCFWEFGDR VKNWITLNEP 180
WSFAFGGYTT GAYAPGRGAA TPEHMKDAIP KHRCHCLHQV GDCQAGDPGT EPYLVAHHLL 240
LAHAEAVKLY REYFQAQGGK IGITLVSQWW EPLNDTPEDK EALERAADFM FGWFMSPVTF 300
GDYPKRMRDI VKSRLPKFSE EESESLRKSY DFLGLNYYTS QYVTDASGTK SDLVSYVTDQ 360
QVKTQVVGPD GKTEIGPQAG SSWLYIYPIG IYKVLKYIKS HYDDPIIYIT ENGVDELNDL 420
KVTVSKGRAD KIRIKYHHDH LEYVKQAMDV DKVKVKGYFI WSLLDNFEWA EGFAVRFGII 480
HVNYKDGYAR YPKESAIWFM KFLNKSIGNE PSQPTKPTKR ALSNGDLEAN SNVEEIPKKK 540
VLKF 544

SEQ ID NO: 3 moltype = AA length = 546
FEATURE Location/Qualifiers
source 1..546
mol_type = protein
organism = Mitragyna speciosa
SEQUENCE: 3
MEIRRSDFGQ NFLFGSATAA YQIEGATEEG GRRPSIWDTM IRRHPDRISN YDDAKVAIDS 60
YHLYKDDVKF MKQIGLDAYR FSISWTRVLP GGRLNAGVNK EGIKYYNNLI NELLAHGIQP 120
FVTLFHFEVP QALEDEYGGF LSDKIVKDFS EFAGLCFWEF GDRVKHWTTL NEPWTFANYG 180
YAVGSFPPNR GSSSADHASV PIVQHRSDRT SAQICENGDP GREPYIVAHN LLLAHAEAVR 240
IYREKFKETQ CGKIGITLVT HWFEPLNDTE DDKEAAKRAF DFKFGWFMEP VTYGRYPKRM 300
TDNVPSDRLK PLTAEESAKL KGSYDFLGLN YYTATYATND PSVKGPPSYL TDWHVKTQTT 360
ATDGKTLIGQ STSSGWIYVY PTGIYKLLSK IKAYYNDPPI FITENGVTDA GDPEGKIANG 420
RVDEIRINYH REHLKEVKHA INENGVDVKG YFLWSLVDNF EWTAGYKDRF GIVYIDYNDP 480
GRTRYPKESA IWFMNFLKKK KDLSTDNEET NGVAAHNTQE LSQSKLQASA KDLPNNEAPA 540
KRARKR 546

SEQ ID NO: 4 moltype = AA length = 502
FEATURE Location/Qualifiers
source 1..502
mol_type = protein
organism = Mitragyna speciosa
SEQUENCE: 4
MAMQVTFLLG ILLLSVHLSN ASSDKAAFNR KSFPDGFVFG ASSAAYQYEG AAFEGGKGPS 60
IWDTFTHEFP GKILNGSNGD VADNFYHLYK DDVKLMKFIG MDAFRFSISW PRVLPRGKLS 120
GGVNKEGVAF YNNLINELIS NGIKPFVTIY HWDTPQTLED EYRSFLSPLI IDDYLDFVEL 180
CFKEFGDRVK HWITMNEPYI FINGGYDGGT QAPGRCSAWL ANDCPAGNSA TEPYIVGHHM 240
LLCHAAAVNL YKQKYQKTQK GEIGITLVSH WVVPYTSSKL DVMATQRALD FMYGWFIHPL 300
IYGDYPRSMR AIVKKRLPTF TTEQVKMVKG SFDFIGLNYY TAKYAANVAT SNSINVSSST 360
DTLANLTSER NGKPIGDPTG VSIFFVYPKG LRDLLAYTKK NYNNPPIYIT ENGIGDANNS 420
TLIKGIKDAQ RVDFYHRHLL AVQEAIKLGV NVKGFFAWSF LDTFEWASGY TLRFGLCYVD 480
YKDGLKRYPK SSALWFKKFL RR 502

SEQ ID NO: 5 moltype = AA length = 530
FEATURE Location/Qualifiers
source 1..530
mol_type = protein
organism = Mitragyna speciosa
SEQUENCE: 5
MATPESHVFP SDSPLKRQDF PEGFIFGASA SAFQYEGAVD QDGRGPSIWD TFLEEKYPDI 60
VKDGGLEAIQ HYYQYKDDVQ VIKETGLDIY RFSISWPRIL PMGRRIVPEV GERMIEDGIN 120

-continued

```
EAGIEFYNGL INKLLANGIE PMVTLYHWDV PQALEDEYLG FLDKRIVCDF ADYADLCFSR  180
FGDRVKYWIT FNEPWSFSVG GYSMGTFAPG RGASSNERAQ MIFDFPTSRA TSKQASMATG  240
APKEGNPGTE PYIVSHHQLL AHAAAVKLYR EKYQREQGGK IGITLVSNWT VPYHDTKEDK  300
DAAQRALDFM FGWFMDPITY GDYPSTMRSL VGNRLPLFTP IESELLVGSF DFLGFNYYTA  360
NYVKNAPNYE SDAHYKLDSQ AECTPVRDNV PIGEQAASDW LYVYPKGILE HVDYIREKYN  420
NPIIYITENG RDELNQPELS YWVSLFDHKR ISYHYRHLQF LKKAIANGAN VEGYFVWSLM  480
DNLEWSSGFK TRFGMNFIDF KNGLKRYPKL SSGWFKFFLH EDVQPKAEEL             530

SEQ ID NO: 6           moltype = AA  length = 533
FEATURE                Location/Qualifiers
source                 1..533
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 6
DDGSDDDDGS SGGQGGSYII RHFNRSSFPT DFIFGAATSA FQIEGGSDRR GTNIWDVFIE  60
NNPELIANKS DAKVAADSYH RFKEDVSLLS GLGFDAYRFS ISWTRLLPTG RISGGINGDG  120
VRYYHALIDE IISKGMIPFA TIFHWDLPQA LEDEYKGFLS EKIVEDFARY SELCFFLFGK  180
KVKHWITLNE PWSYTTLGYA YGAFPPNRCS KSDTLKGSGN SGFSVGRFAS YNSQDGCKYG  240
NSGTEPYIVT HHMLLAHATA VKIYRENYQK DQGGQIGITL VTQWFEPYNP KMKTDVDAAA  300
RSLAFMFGWY MDPLFTGAYP TEMVTYVDGD RLPSFKDHEA KIVKGSYDFI GLNYYTAEYS  360
QDIPCDKSQQ NYLTDPCVNP NKTRNGKPIG GKTGADWLYV YPKGFHELLV YIKDKYNNPP  420
IYVTENGVCR DNRGPRPHDM LRVNYHRGHL MKLRFCLNRK DVNVKGYFVW SLMDNFEWLD  480
GYFVNFGLVR IDYDNKSLNR SCKDSASWFN VFLNKDGGSN EAQICKLFPS SEN         533

SEQ ID NO: 7           moltype = AA  length = 541
FEATURE                Location/Qualifiers
source                 1..541
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 7
MAIKFFVIVS LLAFAHCAFA NTSNCTCDCT GKKITRGSFP PDFVFGAASS AYQTEGAWNV  60
SGKGPSIWDT FTHLYPEKIA FNGTYSSNGD VAIDGYDRYE SDTDIIRDLG LDVYRISISW  120
SRILPKGQIS GGVNPDGIRY YNDVLNALVD KGVEPCVTVF HWDVPQALED LYGGFLSPRI  180
VEDFRNYADV LFKNFGDRVR TWITLNEPWS FSRYAYDLGM LAPGRCTYWD NPDCTGGDSA  240
TEPYLVTHHQ LLAHAATVDL YRKNYQAKYN GLIGITIVSQ WYLPYRAGNP YDELAAQIAL  300
DFMYGWFMDP ITFGRYPLSM QEKVGSRLPQ FTEEESRKLI HSYDFVGVNY YTATFAHFYD  360
EGYKPPYVSY GTDSRVNLTD VDSEGKPCGK GPAGASWLQV YPLGLREILN YTKTVYDNPS  420
IFITENGLCT NTSISKNESI HDDIRTSYLE DHLCCLLEAI DKYKVNVSGY IVWSLFDNFE  480
WASSYEDRFG IYFVDYEDNL TRIEKDSAAW YKGVVAKSPP PPSPILIREI PSISNRKMLP  540
A                                                                  541

SEQ ID NO: 8           moltype = AA  length = 503
FEATURE                Location/Qualifiers
source                 1..503
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 8
MVVLRRGIVL VVLVAVLGLQ TCLGQINRRS FPKGFVFGTA SSAYQYEGAV KVDGRGPTIW  60
DKFAHSFGKV LDFSNADVAV DQYHLYSEDV QLMKDMGMDA YRFSIAWTRL FPNGTGELNQ  120
AGVDHYNNII NTLIASGIEP YVTIYHWDLP QALDDKYSGW LSPQIVNDFA VYAETCFQKF  180
GDRVKHWITV NEPHTFAVQG YDVGLQAPGR CSILLRAFCR AGNSATEPYI VGHNVLLAHS  240
TVVDIYKKKY KPMQKGSIGI TLDTFWYEPA SNSTEDIQAT QRAIDFNLGW FLEPLIVGDY  300
PSSMRTRVGN RLPKFSQAQS LALKGAFDFV GINHYTTWYA KNKSTSIIGV LLNDSLSDSG  360
AITLPFKGLT PIGDRANSIW LYIVPHGIRS LMNYIKQTYG NPLVIITENG MDDGNSPFNS  420
IQDALKDEKR IKYHNDYLTN LLAAIREDRC NVQGYFVWSL LDNWEWGAGF SSRFGLYYVD  480
FNDKLKRYAK DSVKWFKNFL SAA                                          503

SEQ ID NO: 9           moltype = AA  length = 397
FEATURE                Location/Qualifiers
source                 1..397
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 9
PQALEDKYGG FLSSQIVDDF REYVELCFWE FGDRVKHWIT LNEPWSFSVG GYVNGTFAPG  60
RGASSSEQEN DHPAPALLSR CSPWQSQSIS SNGNPGTEPY VVTHNQLLAH AAAVELYKSN  120
FQKSQNGKIG ITLVSQWMEP LDENSKADVE AAKRALDFML GWFMEPLTTG DYPKTMRKLV  180
GSRLPKFSAE QSKQLKGSYD FLGLNYYTAD YVTSASSSTT GGNLSYTTDS QVTHTTDRNG  240
VPIGPQGGSE WLHIYPEGIR KLLVYVKKTY NVPLIYITEN GVDEVNDTSL TLSEARVDNT  300
RIKYIQDHLL NIRLAIGDGV NVKGYFVWSL LDNFEWSEGY TVRFGFIHID YTNNFARYPK  360
DSAIWFMNSF HKEYPKQFLK RTLEDHEDFV SKKRLRQ                           397

SEQ ID NO: 10          moltype = AA  length = 542
FEATURE                Location/Qualifiers
source                 1..542
                       mol_type = protein
                       organism = Mitragyna speciosa
```

-continued

```
SEQUENCE: 10
MALVNCILPM TNASLKPPSL HLSSLGREHQ IFKANVELSS RATPRTKSNS LRINCLKGVA  60
FPEAETISKL INGVLTSSTA STTDTQVISR KDFPTDFKFG CSTSALQTEG SGTEGGRGPS  120
TWDDFIGTGK KAVDSYNRYK EDVKLLKDMG ADCYRFSISW SRLLPDGTIS GGINQEGIKF  180
YNNFIDELIS NGITPFVTLF HFDLPSALQK KYNGFLDSKI VDDFKAYADL CFKTFGDRVK  240
YWITINEPQV FGQYGYKMQL SIPDVVTYPY TAIHNIILSH ATAAKHYQQN YQPTQKGEIG  300
ISLVTEWFEA GSDTKPDKDA SERAFDFLTG WMLEPLVFGD YPFIMKALVR NDLPKFTDEQ  360
KTLIKGSYDF IGVNYYTSRY ATAVSLTLDE KYTSLDQYQH VTLSVYKDGK LIGEKAPGSS  420
EIYVYPPGLG KALAYISKFY NNPKIYITEN GYPEKRDDSI PQATAIKDDA RIQHITAHLQ  480
SVKEAMKLGA NVKGYCMWSL MDCLEMGSAY NVRYGLCYTD YSTLNRYPKK SSEWLKTFVS  540
SK                                                                  542

SEQ ID NO: 11          moltype = AA  length = 521
FEATURE                Location/Qualifiers
source                 1..521
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 11
MKMVSKTSSS RMLLLLLITA FAALSQCAAN AEELEFLNEE KLHFNTPKGL LRKSFPKGFV  60
FGTATSAYQV EGMADKDGRG PSIWDIFIKA PGREPNNATG EVAVDQYHLY KKDVDLMKRL  120
NFDAYRFSIS WPRIFPNGSG KVNWKGVAYY NRLINYMLKQ GITPYANLNH YDLPQALQDR  180
YNGWLGREVV KDFADYADFC FKTFGDRVKN WFTFNEPRVV AALGYDNGFF APGRCSKAFG  240
NCTAGNSATE PYIVAHNLIL CHASAVQRYR EKYQEKQKGK IGILLDFVWY EPHTDTEDDK  300
AAAQRARDFH IGWFMHPLVY GEYPKNLQNI VADRLPKFTK EEVKIVKGSF DYVGVNQYTA  360
YYMYDPHQGK PKDIGYQQDW NCGFTYDRKG VPIGPRAYSY WLYNVPWGLY KAVMYVKENY  420
QNPTIILAEN GMDDPGNLTL PAGLKDTTRI NYYKAYLAEL RKTIDDGANV IGYFAWSLLD  480
NFEWRLGYTS RFGIVYVDFK TLKRYPKMSA LWFQKLLYRQ P                      521

SEQ ID NO: 12          moltype = AA  length = 516
FEATURE                Location/Qualifiers
source                 1..516
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 12
MSSFNLLLVF PVPFFAITSI LSTQYWWVPA WGADDYVRAD FPADFVFGCG TSAYQVEGAA  60
FEDGRTPSIW DTFVHSGLVD EGNGDIACDE YHKYKEDVEL MAEMGLEGYR FSISWSRLIP  120
NGRGPVNPKG LQYYNNLING LISRGIQPYV TLYHVDTPQA LEDEYGGWLS RKIVKDFTAF  180
ADVCFREFGD RVLYWTTINE PNIFALGGYD NGQIPPGRCS FPFGKFCTEG NSSTEPYIAV  240
HNMLLAHSSA VKLYRKKYKA TQGGLLGINI YTFGVVPHTD STADIIATQR TNDFYIGWIV  300
NPLISGDYPD IMKKNVGTRL PVFAPAESKL VKGAFDFFGL NHYTSCFIED DPNSLNRETR  360
DVVADMAVQI QFPGSRFPPD QFPVIPSGLF SVLQYFKEVY GNPRIYVQEN GQRTDRNTTL  420
NDTPRIEYLQ AYIGAVLDAI RNGSNTRGYF LWSFLDGFEL LDGFKSSYGL YYVDLDDKEL  480
KRYPKLSAAW YSNFLKGRSN AMPDRLIDAA NYISVS                            516

SEQ ID NO: 13          moltype = AA  length = 439
FEATURE                Location/Qualifiers
source                 1..439
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 13
MADNSSGKVA IDSYHRHKSD VQLVKNLGLD VYRFSISWSQ ILPGGKLSKG VNQLGIDYYN  60
ALIDELVSNG IKPVVTLFHR DVPQVLEDLY GGFLSPRIVD DFRNYADVCF KYFGDRVKYW  120
VTFNEPWSFS KEGYADGLSA PARCTYWYDS DCAGGDSGTE PYLVTHHLII AHASAVRLYR  180
QKYQSTQNGS IGITNVAEWY LPSTIGSARD EEAVRVQLDF KFGWFLHPIT YGVYPKVMQQ  240
RVGNRLPRFT REQSKWLIGS IDFLGLNYYT GNFATFFDKS YIPKYPSYST DSGVITTGFD  300
SDGTDIGITA GSAWFRRYAV GLKELLNYIK TNYKNPPILI TENGICTMGD TTEDSINDMY  360
RVKYLEEHLC CLLRAIEDDK ANVFGYIVRK MFDALESTQG YDDLSLKRIT KTSTSYWLKN  420
VLSTNDTSLS DHSRTSSGG                                               439

SEQ ID NO: 14          moltype = AA  length = 353
FEATURE                Location/Qualifiers
source                 1..353
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 14
MAGKCAQEEH TVKAFGWAAR EASGALSPYG FSRRATGERD VKINVLYSGM CNFDTEMINN  60
KFGFTKYPYV LGHETVGVVS EVGSKVRKFK VGDKVGVGGL VGSCRKCYSC TNDLENYCTK  120
AELTEAGAGG CSNFVVTDED FVFHWPEKLP LDLGAPLLCA GVTTFSPLKT HGLDKPGMHV  180
GVAGLGGLGH VAVKFLKAFG AKVTVISTSD NKKREAIEKY GADAFLNSSN PEQMRAAAGT  240
LAAIVDTIPS PHSLVPLLDL LLPHGKVIVL GAPSEPFVLP VMPLLQGGRV VAGSSGASLK  300
QIQEMLDFAA EHNIVADAEV IPIDYINTAI KRIEKGDIKY RFVVDIGNTL KSA         353

SEQ ID NO: 15          moltype = AA  length = 360
FEATURE                Location/Qualifiers
source                 1..360
                       mol_type = protein
```

-continued

```
                     organism = Mitragyna speciosa
SEQUENCE: 15
MAKTPEMVHP VRAFGWAARD TSGELSPFKF SRRATGEHDV QFKVLYCGIC HSDLHMIKNE  60
WGMTVYPIVP GHEIVGVVTE VGSKVEKVKV GDKVGVGCLV GSCGNCDMCS QDLENYCPKQ  120
ILTYSAIDED GSITYGGYSD VMVANEHFII RWPENLPLDA GAPLLCAGIT TYSPLRYFGL  180
DKPGMHVGIV GLGGLGHVAV KFAKAFGAKV TVISTSENKK KEAIEKLGAD AFLNSRDPQQ  240
MQAAVGTFDG ILDTVSAVHP IVPLINLVKS QGKVIMVGAP EKPLELATFP LLVGRKTLAG  300
SAIGGLKETQ EMIDFAAKHN ILAEVEVIPM NYVNTAMERL LKADVKYRFV IDIGNSLKSA  360

SEQ ID NO: 16          moltype = AA  length = 363
FEATURE                Location/Qualifiers
source                 1..363
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 16
MAGKSPEGQH PVKTYGWAAR DSSGILSPFK FSRRATGDDD VRFKVLYCGI CHTDLHAVRN  60
DWGFSTYPLV PGHEIVGIAT DVGSKVTKVK VGDKVAVGCF VGSCGKCENC TNDLESYCPG  120
MITTYVGTYH DGTPTYGGYS NEMIASEHFI VRFPEKLPLA GGAPLLCAGI TVYSPMKYYG  180
FDKPGTHLGI VGLGGLGHVA VKFAKAFGAK VTVISTSPSK KDEAINHLGA DAFLVSRDAE  240
QMKASSGTFD GIIDTVAAVH PISPLIDLLK NHGKLILVGA PDKPLELPVF PLLMGRKTVA  300
SSGAGGLKET QEMVDFAAEH NIVADVEVIP MDYVNTAMER LHKGDVRYRF VIDVGNTLAP  360
PGS                                                                363

SEQ ID NO: 17          moltype = AA  length = 362
FEATURE                Location/Qualifiers
source                 1..362
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 17
MAAEKSPEDL HPVKCFGWAA RDTSGILSPF KFSRRATGEK DVRFKVLYCG VCHSDLHCVK  60
NEWGWTNYPL VPGHEMVGTV TEVGSKVTKC KVGDTVGMGC LCGSCRTCEQ CKNDVETYCA  120
ELKMTYLSIY YDGTPTQGGY SNEMVADEHF IINWPENFPL AAGAPLLCAG ATVYSPMKYY  180
GLDKPGTHLG IVGLGGLGHV AVKFAKAFGL KVTVISTSPS KRDEAINQLG ADDFLVSRDP  240
DAMKAAMGTM DNIFDAVSAV HPLMPLFDLL KPHGKLVMVG APTKPYEIVV PTLVMSRKMV  300
GFAAAAGLKE TQEMINFAAE HNITADIELV PMDYVNKAIE RLAKNDVRYR FVLDIANTLK  360
EE                                                                 362

SEQ ID NO: 18          moltype = AA  length = 343
FEATURE                Location/Qualifiers
source                 1..343
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 18
MAGENRSNVF GLAARKISLE IEISGENRVK TCGWAARDSS GVFSPFDFTR RATGEEDVRF  60
KILYCGICHS DLHHAKDEFG FGIYPLVPGH EIVGIVTEVG SKVTKFKAGD KVGVGTIVGS  120
CRSCNRCSND LEPYCPKMMM TYNSIDEDGI PTRGGFSNEM VVNEHFAVRW PDNLPLDGGA  180
PLLCAGSTVY SPMVYYGLSK PGTHLGVAGL GGLGHVAVKF AKAFGVEVTV ISTSPSKKEE  240
AINHLGADHF LVSTDQDQMQ AAMGTLDGII DTISAVHPIM PLFGLLKHDA KLIVVGAPNK  300
PLEVDIPVLI IGRKMLGASA VGGMKETQEM IEFAAKHNIT ADI                    343

SEQ ID NO: 19          moltype = AA  length = 361
FEATURE                Location/Qualifiers
source                 1..361
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 19
MAQTTPNHTQ SVSGWAAHDS SGKITPYVFK RRENGINDVT IEVLYCGMCH TDIHHVKNDW  60
GITMYPVVPG HEITGIITKV GSNVTKFKIG DKVGVGCLAA TCLECEFCKD SQENYCDQVQ  120
FTYNGIFWDG SITYGGYSKM LVADHRYVVQ VPDSLPMDAA APLLCAGITV YCPLKDHNLL  180
EPGVAAKKIG VVGLGGLGHV AVKFAKAFGH HVTVISTSPS KEKEAKERLG ADDFIISTDM  240
KQMQAKKRTL DFIVNTVSAK HSLGLLLELL KVKGTLVIVG APDKPLDLPS FPLIFGKRVV  300
KGGIIGSIKE TQEMMDLCGK RNITCDIELV QTDQINEALE RLAKNDVKYR FVIDIAGKSS  360
T                                                                  361

SEQ ID NO: 20          moltype = AA  length = 357
FEATURE                Location/Qualifiers
source                 1..357
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 20
MGSLDVERKT VGWAARDPSG VLSPYTYPLR NTGPEDVYIK VICCGICHTD IHQVKNDLGM  60
SNYPMVPGHE VVGEVVEVGS DVTKFRVGEC VGVGAIVGCC GNCRPCTADI EQYCNKKIWS  120
YNDVYTDGKP TQGGFAGAMV VDQKFVVKIP DGLSPEQAAP LLCAGVTVYS PLNHFGLKES  180
GLRGAIVGLG GVGHMGVKIA KAMGHHVTVI SSSDKKRGEA IDHLGADDYL VSSDTTRMQQ  240
AADSLDYIID TVPVFHPLEP YLSLLKVDGK LILMGVINTP LQFVTPMVML GRKTITGSFI  300
GSMKETEEML EFCKEKNLAS MIEVVKMDYV NTALERLEKN DVRYRFVVDV AGSKLEQ     357
```

-continued

```
SEQ ID NO: 21          moltype = AA  length = 355
FEATURE                Location/Qualifiers
source                 1..355
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 21
MTSASDKEDC LGWAARDPSG FLSPYKFSRR AIGSEDVSLD IAYCGICYAD VAWTRNKMGH  60
SKYPLVPGHE ITGVVRQVGS QVKRFKIGDR VGVGTYVNSC RECEYCDEGL EVHCSKGQVF  120
TFDSIDVDGT VTKGGYSSYI VVHERYCYRI PDNYPLELAA PLLCAGISVY TPMMRYNMNQ  180
PGKSLGVIGL GGLGHLAVKF GKAFGLKVTV FSTSISKKEE AVNLLGADNF VVSSDEQQMS  240
ALAKSLDFII NTASGDIPFD VYLSLLKTAG VLVLVGFPSE VKFSPVNLIL GMKIVTGSVT  300
GGTKQTQEML DFCASHKIYP EIEIVPIQYS NEALERMIKK EVKYRFVIDI AHSLK       355

SEQ ID NO: 22          moltype = AA  length = 309
FEATURE                Location/Qualifiers
source                 1..309
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 22
MKSNQIYFEF VVAFSPGLCR HEMVGTVTEV GSKVTKFAVG DTVGVGCLVG SCGSCDDCCN  60
DLENYCPNMI LTYLSKYHDG TPTQGGYSNE MVADEHFVIR IPENISLDAC APLLCAGITV  120
YSPMKYYGLA KPGVHIGIVG LGGLGHVAVK FAKAFGAKVT VISTSPSKKE EAINQLGADS  180
FLVSRDPEQM KAATGTMDGI LDTVSAVHPI MPLIDLLKSS GRLLMVGAPD KPLELLVFPL  240
LLGRKGVSGS SIGGIKETQE MIDFAAKHNI TAEIELIPMD YVNTAMERLA KNDVRYRFVI  300
DIANTLVVP                                                          309

SEQ ID NO: 23          moltype = AA  length = 506
FEATURE                Location/Qualifiers
source                 1..506
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 23
MEYIFLLCAL LFSLMLTKIV IKSKNNNSWK KLPPGPRQVP IIGNMHQLYG PLPHHKLRDL  60
AKKYGPLMHL QLGEISTIVV SSPEFAREVY RTHDIIFAER AELLSISIVS YNFLGIVFAP  120
YADYWRQLRK ICTMELLSPK RVQTFRSIRE EEVLNMIKSI ALQKGSSVNL SRKIFSLTYG  180
ITAQAAFGQK GKYQEQFIML VDEMVKLAGG FSLVDMYPSI KLLERISGTR TKLEKLHKQI  240
DGILQTIVNE HKKKLEKEEL GSGEKKVDLV DVLLNIQKRG EFEPPLADTN VKAIIYDIFS  300
AGSETSSTTV EWAISEMIKS PAVMKRAQDE VRNIFKERRN VDESNLDELK YLQAVIKETM  360
RLHPSAPLLL PKESTEKCEI KGYEIPAKTR VIVNAWALGR DPMYWSEAEK FNPERFIDSE  420
LDFRGNHFEY IPFGAGRRIC PGISFALPNI ELPLAQLLFH FDWKLPGDLK QEQLDMTEVF  480
GVTVRRKNDL QVIPVPYSHS SLNKDL                                       506

SEQ ID NO: 24          moltype = AA  length = 506
FEATURE                Location/Qualifiers
source                 1..506
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 24
MELFTILCAF FLFLFMLVKF LNPDRVKVPA QTLPPGPTPL PIIGNLHQLL GSNTHHILKE  60
LASQYGPLMH LQLGQVSTLI VTSPEVAEEF MKTRDIIFAN RPQLFAGRIL RYNCTNIGFS  120
PFNDYWRQLR KICTMELLSS KRVQSFQSIR EEEVLNLVES ISSNWGSTIN LSRRICSLAY  180
GITARAAFGK RSKYQEEFIS LIEEVVKIAA GFTIADMYPS VKIIQIISEM SPKIKRLHKN  240
IDAILENIVD DHKEKNTKTR GTDEEAKDDL VDVLLNVQKS GEFGAPLTNN NIKAVILDIF  300
SAGSESVSTT VEWTMSEMLK NPDIMKRAQA EVREVFGRRR NVDESSLHEL KYLQAIIRET  360
LRLHPPGPLL VPRESSEQCE VNGYIVPVKS RVIVNAWAIG RDSKYWTEAE KFIPERFLDS  420
QVDYHGKDFK YIPFGAGRRI CPGISYALAN VELPLAQLLF HFDWKLPNEL KLEQLDMTEA  480
GFALTVRRKN DLYLDPIPYS RSIFSN                                       506

SEQ ID NO: 25          moltype = AA  length = 398
FEATURE                Location/Qualifiers
source                 1..398
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 25
PIIGNMHQLH GSPIHHLLRD LAKKYGPLMH LKLGEISAII VTSPEMAKEI YKTNDLLFAS  60
RPSPHLAFKI ISYNFVDIIF SAYGNHWRQL RKICTAELLS PKRVQTFKSI REDEVFNLIQ  120
SISSQKGSII NLSRCIFPLS YSITSRIAFG KRNKYTERFI QLVEEINKLV SGFSLADMYP  180
SVKLLQVMSI MRYKLEIAHR QVDQILENIL NDHKQKITDA KQEGGERKED LVDVLLSIQE  240
GGDSEPELTD TAIKAVIFDI FSAGSETSST AIEWALSELI RNPEVIKRAQ DEVRNVHDER  300
GNVDESSLHE LKYLHAIIKE TLRLHPSAPL LLPRECGEQC EINGYVIPAK ARILVNAWAI  360
GRDPMYWTEA EKFNPERFLD SEIDFKGNNF EYIPFGAG                          398

SEQ ID NO: 26          moltype = AA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
```

-continued

```
                     mol_type = protein
                     organism = Mitragyna speciosa
SEQUENCE: 26
MELFLTLCAL LAFFSIALKI VRKTKLVKLP PGPRPLPIIG NLHQLNGSLP HRILTDLAKK  60
YGPLMHLQLG EVSTVIVSSP AVAKEVMKDH DIIFCYRPLL LSPAILNYGG TDIAFAPYGE  120
YWRQLRKICT IELLSSKRVQ TFRSIREQEV SNLFRSIKSA EGSVVNLSKM IYSLTYSITA  180
QAAFGKESKY QDKFISVMQD AIKLMGGFNI ADVYPSIKIL EKITGIRHKL EETHNQIDEI  240
VQNILIEHKA KRAESKLGNR GAKEDLVDVL LNIQSSGEFE PPLTDDNLKA VIYDIFSAGG  300
ETSSTTVIWA MSEMIRNPKV MKKAQDEVRE VFAGNGNVDE SRLHELKYLQ AVIKETMRLN  360
PAVPLLLPRE CSEHCQINGY DVPAKTRVIV NAWAIGRDPK QWTEPEKFKP DRFLDSEIDF  420
RGKTFSFIPF GAGRRICPGI SFALPLVELP LAQLLYHFDW KLPGQSKPDE LDMTEIFGLT  480
VGRKHDLNLV PIPYRPCSVL                                              500

SEQ ID NO: 27          moltype = AA  length = 442
FEATURE                Location/Qualifiers
source                 1..442
                     mol_type = protein
                     organism = Mitragyna speciosa
SEQUENCE: 27
MNHLEIYICY CYHLLSSIFS PHGGHPLFCA IIFLFLALKL LKRSTKKSAL HLPPGPKPLP  60
IIGNMHQLFG SLIHRILRDM AETYGPLMHL RLGEVSTLVV TSPEVAQELL KTHDIIFANR  120
PQLLCPRMFN YGCTDIAFSP YGEFWRQLRK ICVTELLSPK RVQSFRPVRE EEVLKFIRMV  180
FSKEGSTVNL SRMLFSFMYG LTAQVAFGKK NKNQEEFISH IEEATKMTSG FNIADVYPSV  240
KFIQVLSEMS PKMKKMQKNI DKILENIVDE HTNKNQQTEP TDGELKEDLV DVLLNIQKSG  300
DLKIPLNRNN VKSVIMDVFT AGSETSSTVL EWAMSELIRN PGVMKKAQEE VRNAFDGSGN  360
VCESGLPELK YLQAVIKETL RLRPPVPLLI PRESSEQCKI NEFDIPARTR VMVNVWAINR  420
DHRYWIEAEK FDPERFLNCT ID                                            442

SEQ ID NO: 28          moltype = AA  length = 508
FEATURE                Location/Qualifiers
source                 1..508
                     mol_type = protein
                     organism = Mitragyna speciosa
SEQUENCE: 28
MTYFHFNFIT LFIFVAFIMS LVKKWKRSEQ LKKLPPGPWK LPVIGNIHHL IGPPPHHILK  60
NLAHKHGALM HLQLGEISAI VISSPRLAEE IMKTHDLAFA NRPELLVSKI LCYNSSDIVF  120
SPYGDYWRQL RKICTLELLS AKNVRSFGSI RQDEAFHLIS SIRSLAANGT PINLTKTISS  180
FTSSIICRAA FGKVCRDDQD KFIQIMKDSI ALTGGFDISD LFPSWKILHG LSRMKPKLLM  240
VQHEFDTILD SIINQHIENL EKTRNDGDPG DEDLIDVLLR VKKSGDLQFP LTKSNIKAVV  300
YDMFTGGTET SSTTVEWAMA EMIRNPNVMA KAQNEVRKAF MAKKTIEESD IQELRYLKLV  360
IKETLRLHPP VPLLVPRECR EECEIDGYVI PVKTRALINA WAIGRDPEYW SDPDIFQPER  420
FEQNSCEFTG VNFKYLPFGA GRRICPGLQF GLANTELPLA LLLYHFDWKL PDSMNSCCLD  480
MTETCGMTAA RKNNLILAAT SYDPHDTS                                      508

SEQ ID NO: 29          moltype = AA  length = 453
FEATURE                Location/Qualifiers
source                 1..453
                     mol_type = protein
                     organism = Mitragyna speciosa
SEQUENCE: 29
VELQNCPSNF ISFFLFAAIL LTIIKEWRTS KKLKTIQKLP PGPWNLPFVG CMHHLIGGLP  60
HHSLKNLAQK HGPLMHLQLG EVSAIVVSSP NVAKEIMKTH DISFASRPEL LALKNMFYGS  120
IHIAFSPYGD YWRQMRKIRI LELLGAKNVR SFGNIWQDEV LHLVEAVRNL ACKKVNLTEQ  180
IFSYASSMVC RAAFGQVSKD DKSEFAQLMK QVAALTGGFD IADLFPSYKI LHVLSGMKPK  240
LSKVHQKIDV MFENLIKEHD ENRNRRNKGG AEFGQEHLID VLLRIRDSGD LQIPITNNNI  300
KAIIFDMFAA GTETSSSTVE WAMSEMIRNP NVMAKAQNEI RQAFMGKQKI DEADIQGLRY  360
LKLVIKETLR LHPPFPLLLP RECREQCELN GYTIPVKTKV MVNAWAIGRD PEYWRDPEIF  420
EPERLEHNLV DFTGNHFEFL PFGAGRRICP GMT                                453

SEQ ID NO: 30          moltype = AA  length = 514
FEATURE                Location/Qualifiers
source                 1..514
                     mol_type = protein
                     organism = Mitragyna speciosa
SEQUENCE: 30
ASFQMEMHLS FFTLTALFLL LSFLFLLIKE IKGQKNTRDA KKLPPGPKQL PFIGNLHQMV  60
GSLPHHKLRK LADKYGPIMQ LQLGELSTIV ISSTELAKEV LQTHSAAFAN RPQLMVAKIM  120
LYNSLGVTFS PYGDYWRKMR HMYITELLNT KSVHSLFGIL EDEVSKLTMS IESLAGKPII  180
LGEMILSYLN TAISRATVGR ICKDQDALIM ATREAASLAG IFNLADLFPS VKILQLVNRS  240
KAKLEETYRK LDGILEDIIS GHEKNKIGTK ADNQNPAEED IVDILLRLKK SNDFQLPITS  300
SNIKAIIFEL AVAGTITSLV AIEWAISELM KNPKVMEKAQ AEVRQAFKGQ KKISGVDIQK  360
LKYLKWVIKE TLRLHPPGPL LAPRECREQC EISGYTIPIN TVTIINAGAI GRDPKYWNNP  420
ENFEPERFSN SSIDFTGNSF ELLPFGAGKR LCPGINYAVA NIEILLANLL YHFDWKLPGG  480
ITPEEFDMTE KFAAAVTRKD DLCLLAKSYV PSDV                               514

SEQ ID NO: 31          moltype = AA  length = 507
```

-continued

```
FEATURE                 Location/Qualifiers
source                  1..507
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 31
MELRFIPFSL SSINLFLLLS FLFLLVKLWK NFKTSNIAQK LPPGPPKLPL IGNIHNLIGG  60
LPHHVFRDLA KIYGPLMHLQ IGEVLTVVLS SAKLAKDVLV THHPSFASRP SNLAIKIIWY  120
DQEGMGFTSY GEHWNQMRKI CMLELLSIKS VASFKPIRQD EISKLVGSIQ STLGVPVNLT  180
EKIFAYTSSV ICRAAFGRVC KDKEKMIKNL KIALALAGGF NLADVFPSLR LLPIITGFKQ  240
KLIKIHHEMD EILCDVVKEH IANHESGKKG NAETGDQDLI DVLLLQQKTS IQIPITVQNI  300
KAVIIDIFTA GTDALSDTTE WAMAELMRHP RVMAKAQEEV RQLFKGKKTI EEADVQKLKY  360
LKMVIQETLR LHPPAPLIPR ASRENREVNG YMIPDKVQVL VNVWAIGRDP EYWDDPEEFI  420
PERFEQKSVD YLGTQSEYIP FGSGRRMCPG ATFGLASVEL SLANLLYHFD WKLPNGMEPC  480
DLDMDETAGV TAGRKNSLYL VAIACDP                                      507

SEQ ID NO: 32           moltype = AA  length = 260
FEATURE                 Location/Qualifiers
source                  1..260
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 32
MFIVCNIRSD SDMEIELLAL FLTFFSMLLL LLFGSRFRTN NSISHLPPGP WKLPLIGNIH  60
QIVGSPPNRA LRDLAKKYGP VMQLQLGEVS TIVVSSAETA KEVMRTHDII FASRPHIIAT  120
KILSYDSTNI GFSPYGEYWR QLRKICVSEL LGASRVQSFR YIREEETLDL IRWKACNAGS  180
MINLTQKIRS FTYATISRAA FGKKSAEQET FINTVTDCAK LAGGFNIADL YPSVKFLDVI  240
SGLKTKLRRL HRQDTWEHHQ                                              260

SEQ ID NO: 33           moltype = AA  length = 307
FEATURE                 Location/Qualifiers
source                  1..307
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 33
MELIPLPTSF ITSILLLFLF LKLLPKFTTS SKKVGSPKLP PGPWKLPIIG SMHHLAGSLP  60
HRALAGLAKK HGQIMHLQLG EISTVVISGA RMAKEVLKTH DVALANRPKF LAAEIITYNY  120
QDVAMAPYGD YWRQMRKICV LELLSAKNIR ASRFIREDEV WKMVHSIQSS SSSMSRKVYL  180
SEKVTALTNA VTCRASFGSR CKYQDELVKL INEIMEVSTG FDIADLFPTK KFLQIISGTI  240
SRLKKLQSKL DPIFEDIIQE HKEMNPAARG QSEDPENLVD VLLKVNERGG LDFPISTSSI  300
KAIFVVS                                                            307

SEQ ID NO: 34           moltype = AA  length = 511
FEATURE                 Location/Qualifiers
source                  1..511
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 34
MALFEWEKEG FPSPYLLIFA SVSLIVVLKF VSKPKNNRNS KLPPCPPKLP IIGNLHLLGN  60
MPQESLHQLA KKYGPVIFLQ LGEVPTIVIS SARLAKEAMK TFDLALSSRP QIFSAKHLFY  120
NCTDVVFSPY GAYWRNIRKI CILELLSAKR VQSFSYVREE EVAHLVQRVA EAYPGTTNLT  180
KMLGLYANNL VCRVAFGKDF SQGGDNHRHN FQKLLEEYQV LLGGFSIGDF FPSMEFVHKF  240
TGMESRLKDT FRRFDQFFDE ILQEHRNPES RKEHKDLVDV LLEIQKNGDP ETPLTTDNIK  300
SIILDMFAAG TDTNFITLDW AMTELMTNPG VLKRAQAEVR NIVGERRSVS ENDLPQLPYV  360
KAVVKETFRL HPPAPVLVPR ESMEDVTIDG YDIPAKTRFF INAWAIGRDP KSWENPDVFE  420
PERFMNSHID FKGQDYELIP FGAGRRSCPA ITFSAVTVEL ALAQLLHSFD WELPPGVEAK  480
DLDMTEVFGI TMHKIYPLMV VAKPYFSQGL N                                  511

SEQ ID NO: 35           moltype = AA  length = 533
FEATURE                 Location/Qualifiers
source                  1..533
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 35
MERGKIQGKL RSMASSSGLV SDLKFEESIL HPIFFSLFPL IILLFLFKFL SNTRKNQPPS  60
PPGFPIIGNL HQLSSLPHYP LHSLAKRYGP IMLLKFGSVP TVVISSADAA SLIMKTHDLI  120
FSDRPFSRTA NKLLYNMKDI SIAPYGEYWR QLKSICVLQL LSHKKVQAFR NIREEETSVM  180
MDKIKEASLN LTPVNLSEML FSLTNDIVCR SAFGRKYGEG ESGKKFKLLL SEFLKLLNTP  240
SLADFVPWLG WVNRVNGFDA SIDRVAREID EFLESVLHER LVATLGNDSR SSDETIVCES  300
KEDFLDILLN IYKENAAGVT MDRESVKAII LDVFSAGTDT TATVAEWAMT EILRHPIVKE  360
RLQNEIKGVV GSNRDITEYD LEKMPYLKAV IKETLRLHPP IPLLVPRKAR EDVRIMGYDI  420
AAGTMVIINA WAIGRDPAFW EEPEKFKPQR FLDSSIDFKG HDFQLIPFGA GRRGCPGIAF  480
AIASNELVLA NLVRKFDWRF PDGTKGEELD MTECPGVAVH RKIPLLAIPT PLS         533

SEQ ID NO: 36           moltype = AA  length = 573
FEATURE                 Location/Qualifiers
source                  1..573
                        mol_type = protein
```

-continued

```
                     organism = Mitragyna speciosa
SEQUENCE: 36
MAEVIFSSFL TTFCSFSTIT IIFSPFLTST KTRIFSISHT FDLHIEQDIH QKQNKMDNFT  60
PEILCTIFLS FAAFCFYYYY IIHRRTRSPP VTLPPSPSKL PIIGHLHLLT DMPHLRFSQL  120
AQKLGPIIYL QLGQVPTVVI SSAKLAELVL KTHDHVMANR PQIIAAQYLS FGCSDVTFSP  180
YGPYWRQARK VCVTELLSSK RVNSFRLIRD EEVNRMLSTV SAHANSELDV SELFFQLAND  240
ILCRVAFGKR FMDENNEETG KKNELVGVLT ETQALLAGFC IGDFFPSWKW VSSVSGTKRR  300
LMNNLKDLRM VCDEIINEHV RRREHRGPSA DVLDKEDFVD VLLRVQKQDD LEVPITDDNL  360
KALVLDMFVA GTDTTSATLE WTMTELARHP KIMKRAQDEV RQIAASKGKV EESDLQHLHY  420
MKAVIKETMR LHPPVPLLVP RESMAKCTID KYEIPANTRV LINTYAIGRD AESWEDPLEY  480
NPERFLGKDI DFKGQDFRFL PFGGGRRGCP GFSFGLASVE ISLARLLYRF DWALPEGVGA  540
NDVDLTEIFG LATRKRSALK LIPTFNNQHA RVI                               573

SEQ ID NO: 37          moltype = AA  length = 514
FEATURE                Location/Qualifiers
source                 1..514
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 37
MAKQIFVLDL PFSSLFPLAL LLLTLVIWFY ASSRPQKRLP PSPPKLPIVG HLFRLGIYPH  60
RTLQRLSKKY GPLMMIHLGS KPILVASSAD VACEIMKNHD LVFANRPKRS INSKLLYDCK  120
DIATAQYGEY WRQIRSICVL QLLSNKRVQS FRHVREEETS MMIQKIRERC SSSSSSLINL  180
SDALITLTND VICRVALGRK YSNDENGRKS MKIMKEFGEL LGTFSVGDYL PWLAWVNRFN  240
GLDAKLEKVA KHIDQFLEGV IEEHKFRKKG EDYVDSTEKK ALDLVDILLD IQKDNSVGLA  300
LELDSVKAII LDMFAGGTDT SYTAMEWAMS ELLRDPKTLE KLQVEVRKVA QGKSEITEDD  360
LDKMHYLKAV IKETLRLHTP IPLLVPREST QHVKILGYDI PAGTEVMVNA WAIARDPVQW  420
KKPEEFLPER FLNSTIDFRG FNFEFIPFGA GRRGCPGTTF AIAVNELALA KLLHEFDFAL  480
PDGAKPEDLD MSDATGFTVH RKFPLLAVAI PHSC                              514

SEQ ID NO: 38          moltype = AA  length = 501
FEATURE                Location/Qualifiers
source                 1..501
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 38
MGTLIVLFSI LPIVILIILQ NASKKPSKKI HPPPPGPPGL PLIGNLHQFD SSTPHRYLWQ  60
LSRKYGPLMS LKLGSVPVLV VSSAKMAEEV MKNHDLIFCS RPPMHGQRKL SYNGLDIAFA  120
RYGEEWREMR KVAVLHLFSN RRVQSFRSIH EDEVYRMTQK ISTESASSQV TNLSRTVLSL  180
TSTMICRIAF GKRYDEKSLE RIRFYDLLHE VQSAFVGFYF SDYFPSLGWV DYCTGMLSRL  240
EKTFNNLDLF YEELIEEHLH PNRPKSMQGD IIDLMLHLRN DKSTPFDLTM DHIKAMLMDM  300
FTAGTDTVGA AIIWAMTALI KSPRAMKKAQ AEIREVIGNK KVVGEDAAQK LPYLKAIIKE  360
TFRLYPPAPL LVPRQTMENC VIDGYEIKAK TIVYINAWAI ARDPKYWENP DEFLPERFLN  420
STVDLKGQDF QVIPFGAGRR GCPGYSLGLA TVELALANLL YSFDWELPCG MRKEDIDVDV  480
LPGLTMLKKN DLQLVAKNVE V                                            501

SEQ ID NO: 39          moltype = AA  length = 370
FEATURE                Location/Qualifiers
source                 1..370
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 39
MLLQLGSVPT LVVSSADMAR EIFKRHDPAF SGRPVLYVAK KLSFNCSALA FAPYGEYWRQ  60
VRRILVLQLL STKRVQSFEA IRADEVALMV KFIARCSNNN PVNLSKLAFK LANNIICRVA  120
FGRSDDGNAK FREILYETQA LLGESNIADF FPRMAWFNKF NGIESRLEKN FRELDGFFNE  180
VIKVHLDPGK FKPEHEDLVD VLLRSPRDPT QTISLSNDDI KGVLTDILVA GSDTSAATIE  240
WTMAELIRNP TAKRKAQQEV REVMKAKGKV EESDLSRLKY LKLVMKESLR LHPPAPLLVP  300
RETLESCKIG GYQIPAKTRV LINAASIATD PTCWENPLEF KPERFLNSSV DFRGQNFELL  360
PFGAGRRSCP                                                         370

SEQ ID NO: 40          moltype = AA  length = 512
FEATURE                Location/Qualifiers
source                 1..512
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 40
MSTLTFIFIT SLLAFLLYSL LKKILNPTKP LPPGPKPWPI LGNLPHLGTK PHHSLAAIAK  60
TYGPLMHLRL GFVHVVVAAS ASVAAQFLKT HDANFSSRPP NSGAKHIAYN YQDLVFAPYG  120
PRWRMLRKIC SVHLFSAKAL DDFHLVRQGE VSILTRALAS AGNNPVNLGQ LLNVCTTNAL  180
GRVMLGRRVF GDGSGGGDPK ADEFKSMVVE LMVLAGVFNL GDFIPVLDWL DLQGVAAKMK  240
KLHARFDAFL NTILEEHKIN GSDGDKGHVD LLSTLISLKD NADGEGGTLS DTEIKALLLD  300
LFTAGTDTSS STVEWAIAEL IRHPKLLAQA QQELDTVVGP NRLVTEADLP QLTYLQAIIK  360
ETFRLHPSTP LSLPRIAAES CEINGYFIPK GSTLLVNVWA IARDPDMWTD PLEFKPQRFL  420
PGGEKHNVDV RGNDFEVIPF GAGRRICAGM SLGLRMVQLL TATLVHAFDW DLPNGQTAEK  480
LNMEEAYGLT LQRASPLMVH PRPRLAPHAY RT                                512

SEQ ID NO: 41          moltype = AA  length = 524
```

-continued

```
FEATURE                 Location/Qualifiers
source                  1..524
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 41
MVYLIWNYIL FQSVDQESKN HHPKVMMPSA WIWPAFTLLA IVSFLNSLLR KNKKKKLPPS  60
PRGLPILGHL HLLGKNPHQD LHKLSKQHGP IMHLRFGFVS NIIVSSPHAA EQFLKTHDLV  120
FASRPPHEAA KYIDFGQRNL SFGQYGSYWR NMRKLCTLNL LSQLKINSFQ SMRREELGLF  180
LKSLKEAALN RDVVDISAEV AALNANMSCL MVFGKKYADN EFDKRGFKAV IKEGMQLSAT  240
PNLGDYYPYL GVLDLQGLTR RMKAIGKVFD EFFEKIIDEH EQSENQARQA DDFVYTMMAL  300
MKSGETEFQF DRRHIKAILL DMLAGSMDTS ATVVEWILAE LLKHPRVMKK VQRELEEVVG  360
LNRMVEESDL DSLKYLEMVV KEALRLHPVA PLLIPHASME DCTVDGFYIP KNSRVIINVW  420
TIGRDPNAWS DPEEFIPERF VGSNIDLSGH DFQLLPFGSG RRGCPGMQLG LTVVRLLVAQ  480
LVHCFNWELP NGMSPKELDM TEEFGLVVTR AKHLMAIPAY RLSK                   524

SEQ ID NO: 42           moltype = AA  length = 404
FEATURE                 Location/Qualifiers
source                  1..404
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 42
MEFQFISFPL NFLNLLLLFG FLILLVKIWQ KQNTSKIRLP PGPLKLPLVG NMHNLVGGLP  60
HVVLRDLAGK YGPLIHLQLG EVSMVVVSSP EMAKEVLVNH DPAFADKPEC LATKIMWYDQ  120
QGLIFTPYGD HWRQMRRICM TELLSAKIVR SFSFIRNDEI TKLMESIRSS LGFPINATEL  180
FFNFASSMTC RAAFGGICKD NETLIAYLKE GAILASGFNV ADAFPSLKIL PIISRLKGKL  240
LEMHHKTDLI LDDLINQHKV NHEKGKKGNA ESGDQDLVDV LLRQQENGNL QIPITSRKIK  300
ALLFDIFTAG TDTSSITTEW AMSELMKHPS VMAKAQAEVR QVCNGKKTIE EADIQKLVYL  360
KMVVKETLRL HPPIPLIPRA SNENGEVKGY MIPDKFQVMV NAWA                   404

SEQ ID NO: 43           moltype = AA  length = 373
FEATURE                 Location/Qualifiers
source                  1..373
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 43
MDFTFNFIAF FLSLAFVLTL CKKWKRYKAA EKLPPSPWKL PIIGNMHHLI GSIPHHAFRK  60
LSQKHGPLLH LQLGEISSII VSSPRLAKEV MKTHDLAFAS RAEFLGTKIV RYNCSDIVSS  120
PYGEYWRQMR KICTLEIFSA KSVRSFGSIR KDEAMHLIAA IHDLAAAQEP INLAEKLFSY  180
TSSVLCRVAF GKLSKNQQSA FVQLNNESVL LFSAFDISDL FPSLKILHPF LSMKTKLVKI  240
HHKMDLLLEN IIDQHIDNLA RTKMATGESG IKDLIDVLLR VKENGDLQIP ITKNNIKAVV  300
LDIFSGGTET SSGTVEWAMS EMIRNPTVMA KAQSEIRKAF TGKKTIEETD IQDLRYLKLV  360
IKETLRLHPP VPL                                                     373

SEQ ID NO: 44           moltype = AA  length = 513
FEATURE                 Location/Qualifiers
source                  1..513
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 44
MAEVEGYIHL FLIWLIPTIL LQILFRKRTV SRLPPSPLAL PIIGHLHLLA PIPHQALHKL  60
SNRYGPLIHL YLGSVPCVVA SSPEMAKEFL KTHENSFSNR PKSAVIDYIT YSSQDFSFAP  120
YGPYWKFMKK LCMSELLGGR TLDLLLPVRR DEIKRFMELL ERKAKATEAV DIGAELIRLA  180
NNVVSRMAMS KRCSENEDEA GDVKKLIHEI AELTGMFNLS DFIWFCKNLD LQGFKKRIKD  240
VAERFDAMIE KIIEEHQKTR RMHKQSNDAG QEAKDLLDIL LDVSEDESSD IRLTRENIKA  300
FILDIFAAGT DTSAITVEWA LSELMNHPNI MQKAVQEIDS IIGKNRLVDE LDIKNLPYLQ  360
AVVKETLRLH PTGPIIVRES SEDCEIGSYH IPAKTRLFVN VWAIGRDPKS WENPLEFRPE  420
RFLFEEGSEN KSTLDVRGQH FHFLPFGSGR RGCPGTSLAL QVVQTSLAAM LQCFEWKING  480
KVDMEEGPGL TLPKANPLVC TPVARLNPLP IPK                               513

SEQ ID NO: 45           moltype = AA  length = 511
FEATURE                 Location/Qualifiers
source                  1..511
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 45
MDSFPSNWAA YALVWLATVV IVLLSKHFRK KKLNPPPGPK PWPIIGNLNL IGSLPHRSLH  60
QLSLTYGPLM QLRFGSFPVV VGSSVEMAKV FLKTMDVTFA GRPKTAAGKY TTYNYSDITW  120
SPYGPYWRQA RKMCLTELFS AKRLESYEYI RVEEMNSLIL GLFKSSGKPV LLKDYLSTVS  180
LNVISRMVLG KRYLDESENS IVAPEEFKKM LDELFLLNGV FNIGDSIPWL DFLDLQGYIK  240
RMKILSKKFD RFLEHVLDEH IVRRKDEKNC ASKDMVDVLL DLADDPTLEV KLERHGVKAF  300
TQDLLAGGTE SSAVTVEWAI SELLKNPELF EKATEELDRV IGQNRWVKEK DMPNLPYIEA  360
IVKETMRLHP VAPMLVPRCS REDCKVAGYD IQKGTRVLVN VWTIGRDPQL WEKPAEFHPD  420
RFIGKDIDVK GQDYELLPFG SGRRMCPGYS LGLKVIQSNL ANLLHGYKWK LPNDMKPADL  480
DMEEIFGLST PRKIPLVACV EPRLPRHLYS L                                 511

SEQ ID NO: 46           moltype = AA  length = 275
```

-continued

```
FEATURE                 Location/Qualifiers
source                  1..275
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 46
MEFFFLVCVI IFCFMAVKVV KKSRGKSLQR LPPGPTQLPI IGNIHQLRGS LPHRTLRDLA  60
EKYGPLMHLQ LGELSMIIVS SPQVAEEIFK THDVIFSQRS NLIASSIINY YLVNIIFAPY  120
GNYWRQIRKI CTMELLSPTR VQTFRSIREE EVLNMMKSIS VKEGSVLNLS KMIYSLTYGI  180
TARAAFGKPR QDQEEFKALI EKITEAFGGF SIVDMYPSVK LLQVISGMRF KLKKIHNRVD  240
DIIQNILNEH KEKTSKQEPG SGETKDDLVD VLLNI                             275

SEQ ID NO: 47           moltype = AA  length = 433
FEATURE                 Location/Qualifiers
source                  1..433
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 47
MVLLFLIILL VLIILTHQKN KKVGKRCHPP GPPGLPFIGN LHQFDAEKPH KFLWRLSKKY  60
GPIMSLQLGF VHVVVISSAR IAEEALTTHD LVFSGRPAAV GRRKLSYDRR DIAFSPYSEY  120
WRDMRKICVL HLFSLKRVES FRPIREYETS YMIQKIAKHS SLSQPINLRV LIMSLTSTII  180
CRLAFGKRYD EEGHERKRFD ELLLESQAMI AGFFVADYLP SLSWLDKLFG MMTRLEKNFN  240
ELDLFYKELI EEHLNPNRPD TMKNDLLDLL IQLKQEQSST VGFSWEHIKA ILMDIFVAGT  300
DTSAATIIWA MTALMRSPSA MRRVQVEVRE VVGKKGAVDE EDIHELPYLK AVIKETLRLY  360
PPAPILGPRE TTERCNLHGY PIESKTLVIF NAWAIARDPE CWKNPDDFFP ERFLNSNIDV  420
KGHDFEVIPF GAG                                                     433

SEQ ID NO: 48           moltype = AA  length = 517
FEATURE                 Location/Qualifiers
source                  1..517
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 48
MVKKMGSLVE ALDTETMMLF FIVPLVFLFL VSRFRRKPHP PGPKGWPLIG NMMIMNQLSH  60
RGLAQLADQY GGLFRLKMGF LNMVAVSSPE IARQVLQVQD NIFSNRPATI AISYLTYDRA  120
DMAFAHYGPF WRQMRKLCVM KLFSRRRAES WDSVRDEVDD MIKITASNTG SAVNLGELVF  180
GLTRNIIYRA AFGTSCREGQ DEFIQILQEF SRLFGAFNLA DFIPWLNWLD PQGLNKRLVK  240
ARASLDGFID SIIDDHMEKK KKMSGSVVEG DSDMVDELLA FYSDEAKVSE SEDLQNAIRL  300
TRDNIKAIIM DVMFGGTETV ASAIEWAMAE LMKSPEDLKK VQQELSDVVG LHRKVEESDF  360
DKLTYYKCCL KEVLRLHPPI PLLLHETAED SEVAGYYIPA KSRVMINAWA IGRDKDSWED  420
ANMFKPSRFL KDGVPDYKGS NFEFIPFGSG RRSCPGMQLG LYALEVAVAH LLHCFTWDLP  480
DGMKPSELDM DDIFGLTAPK ASRLVAVPTP RLLCPLY                           517

SEQ ID NO: 49           moltype = AA  length = 508
FEATURE                 Location/Qualifiers
source                  1..508
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 49
MENTWLFLAL ASLVALAFLS KVSSDRRSKL NHPPGPKPWP IVGNLNLLGS IPHQSLHSLS  60
QKYGEIMQLK FGSKPVVVAS SPEMAKEFLK THDHIFASRP TTAAGKYTSY NCSDMTWAPY  120
GPYWRQARKI YLTEIFSPKR LDFFKSIRVE ERNTFISNLF ALSGKPVVMR DHLMRLGQST  180
SCRMVLSNKY VAQSERDGSL VTFEEFQEMI DTWFLLGGVF NIGDWISWLD RFDLQGYVKQ  240
MKELNKKFDR FHNQVIDDHL AKRKTERDFI PKDMVDILLQ LAEDPNLEVK LTSDHIKGLI  300
QDLLAGGIDT SATTVEWAMN ELLKQPSLIG KATEELDRVI GRDRWVEEDD FSKLPFIEAI  360
IKETLRLHPL ATLLAPRYAI KDCTVAGYQI ANGTTVFINT WSIGRNSRYW DSPEEFLPER  420
FLEKDVDMKG QNFALLPFGS GRRRCPGYNL GLKLVRSMLA NLLHGFNWRL PHGMAPEDIC  480
MEELYGLTTH PRISLALIPE PRLNVNLY                                     508

SEQ ID NO: 50           moltype = AA  length = 523
FEATURE                 Location/Qualifiers
source                  1..523
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 50
MSSFQASQEE AKRWLLNDSG VVSLLLPWFV IFSLFISTWW FLTTSKKKNP PPSPPRLPII  60
GNLHQISSLA HRSMKSLADK HGPLMLLHLG SQPVLVATTA DATREVLKTH DVAFSGRPES  120
AVTQSIFRNV KNISLSPYGD SWKHVRSMAV NLFLNQKKVQ SFRSIREEEV SLLVERIKES  180
CASSSVICMN ILLTTFANDT VSRITVGKRY SGQDGLKLAE ILTTGAELLG AYMIWDFFPR  240
LAWIAYLIGL EAKIKKVAKD FDQCLDNLVE EEMNRNKTGG SDSDVGEGDQ GNDNQNLLQA  300
LVELHRTNST GYPLDRKIIK GLVVDMIAGG LDIYSLLEWA MAMLIKHPDV MKKLQDEVKE  360
FAGSKSFVSE GDLSKLQYLK AVIKETFRFC APGAFIPRIA TKDVKVMGYD IAAGTQLLVN  420
KWALGRDPTL WEKPEEFRPE RFLDSSIDLK GQNFELLPFG SGRRVCPGTT FALLIVELAI  480
ANLVCNFNFA LPDGERAEDL DITEAPGVMS RRKTPLLLVP SLR                     523

SEQ ID NO: 51           moltype = AA  length = 523
FEATURE                 Location/Qualifiers
```

-continued

```
source                  1..523
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 51
MLQIFANTFY NGSKFTNDQL LLVIFSYVLC GLALIWFVVM YVKKSNKGQP PLPPGPTALP  60
LVGNLLSLDP QLHVYFTTLS RTYGPILRLW LGKKVAIVIG SPVLAREVLK DQDITFANRD  120
VPASAREATY GGSDIAWTPY GPEWRMLRKV CVREMLSNAT LDSVYSFRRR ELRQTINFMY  180
SQAGKPVNVG EQMFLTVLNV ITNMLWGGTV KGEERASLGA EFRQVVNEMT QCLGMINVSD  240
LYPGLARFDL QGVQKKTKVL AGRFDRIFER MIDKRAKGIA DEDDKESKDF LQFLLKLKDE  300
GDSKTPLTMN HIKALLMDMV VGGTDTTANT VEFALAEMMN KPEILKKVQQ ELETVVGKDR  360
IVEESDIQKL PYLDAVMKEV LRLHPALPLL IPHCPIETCI VGGYTIPKGA RVFLNVWAIH  420
RDPSIWENPT EFRPERFLNS KWDYSGNDFN YFPFGSGRRI CAGTAMAERM FMFSLASLVH  480
SFDWKVPEGE KLDLEEKFGI VLKMRVPLVA IPTPKLSDAS LYE                     523

SEQ ID NO: 52           moltype = AA  length = 511
FEATURE                 Location/Qualifiers
source                  1..511
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 52
MELAQLQFVW GMTLLIPPLL ILLLRRKKSS FNRLPPGPPG WPIFGNMFDL GSMPHKNLAG  60
LKQKYGPIIW LRIGATNTMV VQSSKTAAEL FKNHDIYFVE RTVTEAMKSH GFDKGSLSLA  120
PYGSYWRVMK RIMTVEMSVN KRINETVAIR RKCIDDMVSW IEKEADARKE SRRGIHLARF  180
VFLASFNMLG NLMLSRDLVD PESEEGMEFF SAMVGLMEWV GHPNIVDLFP WLRRLDPQGL  240
RRKAERDMRK TMQIVSKFVK ERLQDRQTGG NRKKDFLEVL LEFEGNGKDE PAKISEHDLN  300
IIIMEVFIAG SETTSGTTEW AMVELLSNPT TMSKLKNELN EVVGRGKKFQ EGDIDNLKYL  360
QAVVKETLRL HPPIPLLVPR RATQDTNFMG YDIPRDTQVF VNVSAIGRDP ECWEDPNSFK  420
PERFLNSNID FKGNHFELIP FGAGRRICAG IPLAYRMLHL VLGSLVHEFD WYVHNSVSDD  480
VRDTKERMGV VVRKTEPLIA MPISKNSANK G                                  511

SEQ ID NO: 53           moltype = AA  length = 501
FEATURE                 Location/Qualifiers
source                  1..501
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 53
MENMALLLDI GEYFLLLPIF FLPLLYFIFK QFKQSFASKT PPLPPGPTPW PVLGNIPHMG  60
KMPHVTLSNF AETYGPLMSL KLGTQYMVVG SSPAAAIEIL KTHDRNLSAR YVPKVLPASQ  120
EELNRSSIGW TEVCNDGWKY LRTLCRSELF SIRALDNQSC LREKKVLDMV EHLRAKEGDV  180
VDIGELAFAT VFNMLSNVLV SRDLIDLDEE RDDGGMKSLV RAIMEVGTAP NISDFYPILC  240
KFDLQGLTKK SIKLGIKVRA VWEPIIEERR QKGVPLSQQD FLDTLLENNF TNVRIHQLLM  300
ELFTAGTDTS TSTIEWTMTE LVKNPASMKK VREELENEIN QDLPKESQVL QLPYLQACIK  360
ETLRLHPPAP LLLPHRAPET CQVMNYTIPK NAQILVNVWA IGRDPSIWDE ASKFKPERFL  420
SCGLDYKGND FEFLPFGAGR RICPGLPMAA RQVPLVIASL IHCFDWSLPE GKDPQQLDMK  480
ESFGFTLQKE QPLLLIPKAR K                                             501

SEQ ID NO: 54           moltype = AA  length = 500
FEATURE                 Location/Qualifiers
source                  1..500
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 54
MELYTLLFCI SLLLFIIGPF LHGSSTKKTP PGPKGLPIIG SLLELGSQPH QSLAALAKIH  60
GSIMTLKLGS VTAIIASSPE TAKEILQKQE QIFSDRTVPD VVTSQPNPQA TLAWVPGDHQ  120
WRNKRRICST QMFTTQRLDL LQHLREKKVE ELVLYFKKRS VLGAPVDIGR AAFATTLNLM  180
SNTIFSIDMV DSEFETAQEF KDLVWRIMED AGKPNLSDYF PVIKWFDLQG VKRHIRPAYR  240
RLHEIFEEVI VERLEDRKLG KRKSRKDDFL DVLLDQCEEE ESGFDRETIK PLILDLFIAG  300
SDTSAITTEW AMAELLRKPE AMHKVRTEII QKIGFERSVK ESDVNQLSYL QAVVKETMRL  360
HPAAPLLLPY KANKDTEIFG FTIPKNSQVL VNAWAIGRDP NHWKNSLSFL PERFFGSELD  420
YKGRDFEYIP FGAGRRICPG MPLAVRMVNL MLASIIQAFS WKLPDGITPE KLDMEEQFGV  480
TLKKAVPLCA IPIIEENLGF                                               500

SEQ ID NO: 55           moltype = AA  length = 364
FEATURE                 Location/Qualifiers
source                  1..364
                        mol_type = protein
                        organism = Mitragyna speciosa
SEQUENCE: 55
MDLARNGDRA GELLQAQAHI WNHIFNFINS MSLKCAVQLG IPDIIEKHGQ PMTLAQLCHA  60
LSISNTKASF LYRLMRVLIH SGFFIKAKIP GNELEEGYEL TSASKLLLKS NPFSVTPFLL  120
AMLAPISTDL WHHFSKWFQS DDETPFHTCH GTSGWEMASQ EPWINQSFNE AMASDARLVS  180
SILIEHCGHV FSGLNSLVDV GGGTGTLAKA IADAFPDLKC TVLDLPYVVD GLDGGRKNLA  240
FIGGDIFEAV PPADAVLLKW TLHDWSDEDC VRILEKCKQA IPSKENGGKV IIIDMVINDQ  300
QKGADDGEHK SIETQLFFDM LMMVDTKGRE RNEKEWAEIF SAAGFKDYKI THALGLRSLI  360
EIYY                                                                364
```

-continued

```
SEQ ID NO: 56          moltype = AA  length = 348
FEATURE                Location/Qualifiers
source                 1..348
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 56
MATLENSAEL LRAQAHVWSQ TFNFKNSASL KCAVQLGIAD VIQKHGKPIT LSKLAAALPI  60
NPSKANYIYR LMRILVNAGF FAQEKEGYSL TSAGRLLLSD EPVNARAFIL VVLDPDTIKP  120
WNFLSDWFQN DDPSPFDTAH GKNFWDFTAE EPKFSKLFNE GMVSDSHLIT MVLVTECKYV  180
FEGLTSLVDV GGGTGTVARS IAEAFPNLKC TVFDLPHVVA NQEGTLNLDF VAGNMLEKVP  240
PANAILLKWI LHDWSDEACV KILKNCKKAI PGRDIGGKVI IIDMVMESQL EDDESTEAQI  300
CLDMQMLVLY RAKERSEKEW AKLFSDAGFP SYKVFPALGT RAIIEVYP               348

SEQ ID NO: 57          moltype = AA  length = 360
FEATURE                Location/Qualifiers
source                 1..360
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 57
MDTNDEGSKL LHGQAEIWKH TFRFADSMAL KCAAELGIAD IIHSHGRPLT LAEIATNITN  60
SSSPHIPYLS RIMRFLVRNK IFTSTDIPSD HGGDSPSTTL YGLTAASRWL LNDNDGLSLA  120
PFLLMENHPW LVSPWHQLSS CVREGGIAFQ KAHGKEIWEL ASENPEFNTI FNDGMGCTGK  180
ITMKAILSGF KDAWNTIESL VDVGGGTGQT IAEIVKVYPH IKGINFDLPH VVATAPKYDG  240
VSHVGGDMFD SIPTAQAVLM KWIMHDWGDE DCVKILKICR KAIPEKTGKI FIVDVVLKPD  300
GDGLFDSIGM IFDLLMIAHS SGGKERTEPE WKKLLEKGGF PRYNITEIPA CFSIIEAYPV  360

SEQ ID NO: 58          moltype = AA  length = 345
FEATURE                Location/Qualifiers
source                 1..345
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 58
MEEAKTREEE AQAQVDIWKY IFGFTEMAVV KCAIELGIAD VLESREGQAM TLGELSSALG  60
CSPSSLYRIM RFLTNRGIFK QLSGTTYAQT PLSCLLVRNG EKSMAAFVLL ESSPVMLAPW  120
HHLRERVSAD GPPAFDAAHG QDVWSYAEKN SAHSKLINDA MSCDARFAVS AIIDGCPQVF  180
EGIGSLVDVG GGDGTALRTL LKACPWIRGI NFDLPHVVSV TSHCDGVEHV GGDMFDSVPK  240
ADAAFIMWVL HDWGDDECIS ILKNCREAVP QDTGKVIIVE AVTDGGGEDK LKDVGLMLDM  300
VMMAHTTRGK ERTSKEWANV LNQAGFSRHT VTRIQAVQSV IEAFP                  345

SEQ ID NO: 59          moltype = AA  length = 352
FEATURE                Location/Qualifiers
source                 1..352
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 59
MGSAEEEASL FAMQLASASV LPMVLKSAIE LDLLELIAKA GPGAYVSPSK LAAQLPTHNP  60
QAPVMLDRIL RLLATYSVLN CKLRDLPDGG VERLYGLAPV CKFLTKNADG VSMAPLLLMN  120
QDKVLMESWY HLKDAVIDGG IPFNKAYGMT AFEYHGTDPR FNKVFNQGMS NHSTITMKKI  180
LEVYKGFEGV KTVVDVGGGT GATLNMIIDK YPTIKGINFE LPHVVEDAPF YPGVEHVGGD  240
MFVSVPEGDA IFMKWICHDW SDDHCRKLLK NCYHALPEGG KVILAECILP DAPDTSLATQ  300
NVVHIDVIML AHNPGGKERT EKEFEALAKG AGFKEFRKIC SAVNTWIMEL CK          352

SEQ ID NO: 60          moltype = AA  length = 373
FEATURE                Location/Qualifiers
source                 1..373
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 60
MEKEDCLAKT NSNENSEEVV KLGGKVEEEE QLFSYAMQLV TSVSLPRVLL AAFRLRVFEI  60
IAEAGPGAQL SASEIAAQVS GKNKDAAAML DRMLRLLASH SVLSCGARSQ GRVYGLAPVA  120
KYFVQSKADG GGTLGSLLAL IQDKVFIDSW YQLEDAVREG GVPFDRVYGT HAFEYPGRDP  180
RFNEVFNKAM LDHTTIAINR ILERYKGFEN LKTLVDVGGG LGVTLNVITT KYPNLKGINF  240
DLPHVIQRAP GYPGVEHIGG DMFESVPEGE AIFMKWILHD WDDGHCLKLL KNCYKALSDN  300
GKVIVVDALV PVIPDTSPCV KTTCQSDLIM MTQNPGGKER SEPEFLALAT AAGFKGIRLE  360
CYVCNLWVME FYK                                                     373

SEQ ID NO: 61          moltype = AA  length = 290
FEATURE                Location/Qualifiers
source                 1..290
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 61
MEISENAQLK SASPSSIENG QEGEKQDHFS HVMQLVTSAS LPMVLYTAVK LNLFEIIAKA  60
GPGAKLSPSE IACRLPTTKN PDAASMLDRM LRLLSSHSLL TYDVVEVAGG AETDIGYERV  120
YGLSPVGEYL IPDEEGNSVA PLVELLLDKV LINTWYELGN AVLEGGIPFN RVHGTHAFDY  180
PSRNPRYNEL FNNGMVGPTA ITMKELLQQY KGFELLQTLV DVGGGLGITL HKIIAKYPSI  240
```

-continued

```
RGINFDLPHV IEDAPPYPGM EHIGGDMFES VPGGDAIFMK MILHDWSDDH            290

SEQ ID NO: 62          moltype = AA  length = 374
FEATURE                Location/Qualifiers
source                 1..374
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 62
MAPTLDTIPT VDVKEMPEEA HPMKGGDDFN SYSQNSCYQR GVIDAAKVVI VEAVNEKLDL  60
KNNANFDPSK PFHIADFGCS TGPNTYFAVQ NIVEAVDQKH KSLHEKIQPE FHVFFNDHVN  120
NDFNILFRSL PQDRDYFAAG VPGSFYTRVF PKASVHFAHC SYALHWLSKV PKEIQDKNSP  180
AFNRGRIHYT GTEKHVVKAY FSQFQKDMGS FLKARAQEIV GGGLVVIQIP GLPSGEVLFS  240
RTGAGMLHAL LGSSLMELVN LGIISEESVD SFNLPQYHPS IEELEMVIEM NNSFTIEKVG  300
ALNHPMKSLP FDVQMTSLQV RAVMEGILSD HFGGKILDQL FEIYTTKLQE NYNVFDKEIR  360
RDADLFLVLK RKSN                                                   374

SEQ ID NO: 63          moltype = AA  length = 247
FEATURE                Location/Qualifiers
source                 1..247
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 63
MAPTVKTPTP IPTPNPNPSL DIKVMPEEAH PMKGGDDLNS YSQNSCYQKG VIDAAKSVII  60
EAVTEKLDLE NNITTFDPAK PFRIADFGCS TGPNTYFAMQ NVVEAVELKY KSLKKDQPEF  120
HVFFNDHVNN DFNILFRSLP QNRDYFAAGV PGSFYTRVFP KASLHFAHCS YALHWLSKVP  180
KEIQDKNSPA FNKGGIHYTG TEKHVVKAYF GQFQKDMDSF LKARALEIVG GGLMVIQIPG  240
LPSGEVL                                                           247

SEQ ID NO: 64          moltype = AA  length = 389
FEATURE                Location/Qualifiers
source                 1..389
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 64
MLQQLPSGFR LQSGAAAEVL AKNRAKMEEG KLVSHTYHME GGDGPYSYAK NSSYQREAVD  60
SIKEMMNELI AQHLDLGNLY SFRVADFGCS VGPNTFSAVQ NIIEAVVNKY KSQGMESQIP  120
EFHVFFNDHA ENDFNTLFRN LPTTRGYFPA GVPGSFHRRL LPSSTLYFAH CSTALHWLSK  180
IPKEVMDKNC PAWNKGRIHY AGAAKDVEEA YSTQFGKDLD SFLSARAQEL VPGGLLTLVT  240
LGFPDGVQVS DSSMGESFNI LGSCFQDMVE EGAIVEELVD SFNLPFYYPS LSELKSLIEV  300
NGLFDIKRLE KLAAPVRREK PDLAVCTLHL RAIIGGLIKE HFGEGIIDNF FERHAKKFVE  360
NPIISDERYF KEASFFVFLK RNTTLQCNH                                   389

SEQ ID NO: 65          moltype = AA  length = 375
FEATURE                Location/Qualifiers
source                 1..375
                       mol_type = protein
                       organism = Mitragyna speciosa
SEQUENCE: 65
RERERDREEM ESVQSNSSSS DQFAMKGGDD DFSYTKNSTH QRDAIQATKF FIQESIAEKL  60
DVNKFCGKAF CVADLGCSVG PNTLIAMQNI VEAVELKFKN RKGFHSPTIP EFQVFFNDHT  120
VNDFNTLFRS LPTGHDKRYY GVGVPGSFYG RLFPCDSIHI MHTSFSTPFL SQVPKEVMDK  180
NSAAWNKGRI HHNYAKADVL KAYEAQHAED IDCFLTARAK ELVHGGLLMD VTSFRPDGVP  240
HTHVLTNIGM EVLGYCLMDL AKKGLIDEEN VDSYNVPVYL QSPEELKQAV QRNKYFSIEK  300
MESVPMMIDS DVSAKAQQYS LGMRAVMGDV IREQFGAEIV DKLFDLFKKK LEEHPNFATA  360
VVLDMFVLLK RNAQD                                                  375
```

1. A polypeptide encoding a strictosidine glucosidase that removes the glucose moiety from strictosidine to produce strictosidine aglycone.

2. The polypeptide of claim **1**, wherein the strictosidine glucosidase is from *Mitragyna speciosa*.

3. The polypeptide of claim **2**, comprising or consisting of a polypeptide variant having at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to the sequence of any one or more of SEQ ID NO: 1-13, or a fragment of any thereof.

4. A polypeptide encoding a geissoschizine synthase that opens the tetrahydropyran ring of strictosidine aglycone to produce geissoschizine.

5. The polypeptide from claim **4**, wherein the geissoschizine synthase is from *Mitragyna speciosa*.

6. A polypeptide encoding a hydroxylase that hydroxylates the indole ring of corynantheidine at the C9 position to produce 9-hydroxycorynantheidine.

7. A polypeptide encoding a hydroxylase that hydroxylates the indole ring of mitragynine at the C7 position to produce 7-hydroxymitragynine.

8. The polypeptide of claim **6**, wherein the hydroxylase is from *Mitragyna speciosa*.

9. A polypeptide encoding an O-methyltransferase that methylates the hydroxyl group at the C9 position of 9-hydroxycorynantheidine to produce mitragynine.

10. The polypeptide of claim **9**, wherein the O-methyltransferase is from *Mitragyna speciosa*.

11. A method for modifying an indole alkaloid, wherein the method comprises contacting the indole alkaloid with the polypeptide of claim **1**.

12. A method for modifying an indole alkaloid, wherein the method comprises contacting the indole alkaloid with the polypeptide of claim **4**.

13. A method for modifying an indole alkaloid, wherein the method comprises contacting the indole alkaloid with the polypeptide of claim **6**.

14. A method for modifying an indole alkaloid, wherein the method comprises contacting the indole alkaloid with the polypeptide of claim **7**.

15. A method for modifying an indole alkaloid, wherein the method comprises contacting the indole alkaloid with the polypeptide of claim **9**.

16. A method of producing strictosidine aglycone, the method comprising carrying out an enzymatic reaction with the polypeptides of claim **1**.

17. A method of producing 4,21-dehydrogeissoschizine or geissoschizine, the method comprising carrying out an enzymatic reaction with the polypeptides of claim **4**.

18. A method of producing 9-hydroxycorynantheidine, the method comprising carrying out an enzymatic reaction with the polypeptides of claim **6**.

19. A method of producing 7-hydroxymitragynine, the method comprising carrying out an enzymatic reaction with the polypeptide of claim **7**.

20. A method of producing mitragynine, the method comprising carrying out an enzymatic reaction with the polypeptide of claim **9**.

* * * * *